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Report No: PAD00070

INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT
ON A
PROPOSED GRANT

IN THE AMOUNT OF SDR 69.5 MILLION
(US\$92.2 MILLION EQUIVALENT)

TO THE

REPUBLIC OF CHAD

FOR A

CHAD DIGITAL TRANSFORMATION PROJECT

September 4, 2024

Digital Transformation Global Practice
Western and Central Africa Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective July 31, 2024)

Currency Unit = Special Drawing
Rights (SDR)

US\$1 = SDR 0.75277397

FISCAL YEAR

January 1 - December 31

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ABBREVIATIONS AND ACRONYMS

ADETIC	<i>Agence de Développement des Technologies de l'Information et de la Communication</i> (ICT Development Agency)
AfDB	African Development Bank
AM	Accountability Mechanism
AMOA	<i>Assistance à Maitrise d'Ouvrage</i> (Implementation Support Technical Assistance)
AML	Anti-Money Laundering
ANATS	<i>Agence Nationale Des Titres Sécurisés</i> (National Secure ID Agency)
ANSICE	<i>Agence Nationale de Sécurité Informatique et de Certification Electronique</i> (National Agency for Computer Security and Electronic Certification)
ARCEP	<i>Autorité de Régulation des Communications Électroniques et des Postes</i> (Regulatory Authority for Electronic Communications and Post)
AU	Africa Union
ASYCUDA	Automated System for Customs Data
BEAC	Bank of the Central African States
CAPEX	Capital Expenditure
CEMAC	Central African Economic and Monetary Community
CEN	Country Engagement Note
CERC	Contingent Emergency Response Component
CERT	Computer Emergency Response Team
CFT	Combating the Financing of Terrorism
CIP	<i>Centre des Incidents de Paiement</i> (Payment Incident Center)
CR	Civil Registration
DA	Designated Account
DE4A	Digital Economy for Africa
DFIL	Disbursement and Financial Letter
DFS	Digital Financial Services
EdTech	Educational Technologies
e-GP	electronic Government Procurement
ENASTIC	<i>Ecole Nationale Supérieure des Technologies de l'Information et de la Communication</i> (National School of ICT)
E&S	Environmental and Social
ESS	Environmental and Social Standard
EU	European Union
FCV	Fragility, Conflict, and Violence
Fintech	Financial Technology
FRR	Financial Rate of Return
G2B	Government-to-Business
G2G	Government-to-Government
G2P	Government-to-Person
GB	Gigabyte
GDP	Gross Domestic Product
GEMS	Geo-enabling Initiative for Monitoring and Supervision

GHG	Greenhouse Gas
GNI	Gross National Income
GoC	Government of Chad
GRS	Grievance Redress Mechanism
GSMA	Global System for Mobile Communications Association
HEI	Higher Education Institution
ICT	Information Communication Technology
ID	Identification
IDA	International Development Association
IDP	Internally Displaced Person
IFC	International Finance Corporation
INSEED	<i>Institut National de la statistique, des Etudes Economiques et Démographiques</i> (National Institute for Statistics, Economic, and Demographic Studies)
IPF	Investment Project Financing
IT	Information Technology
ITU	International Telecommunication Union
IXP	Internet Exchange Point
JRC	Joint Research Center
M&E	Monitoring and Evaluation
MATD	<i>Ministère de l'Administration du territoire et de la Décentralisation</i> (Ministry of Territorial Administration and Decentralization)
MB	Megabyte
MCENDA	<i>Ministère des Communications, de l'Economie Numérique, et de la Digitalisation de l'Administration</i> (Ministry of Communications, Digital Economy, and Digitalization of Administration)
MDA	Ministries, Departments and Agency
MFD	Maximizing Finance for Development
MFBEF	<i>Ministère des Finances, du Budget, et l'Economie, et du Plan</i> (Ministry of Finance, Budget, Economy, and Plan)
NDC	National Data Center
NGO	Non-governmental Organization
NNI	<i>Numéro National d'Identification</i> (National Identification Number)
NPV	Net Present Value
NRA	National Risk Assessment
NREN	National Research and Education Network
OPEX	Operational Expenditure
PCE	Private Capital Enabling
PCM	Private Capital Mobilization
PIM	Project Implementation Manual
PIU	Project Implementation Unit
PKI	Public Key Infrastructure
PND	<i>Programme National de Développement</i> (National Development Program)
PPP	Public-private Partnership
PPSD	Project Procurement Strategy for Development

PSC	Project Steering Committee
PwD	Person with Disabilities
P2G	Person-to-Government
P2P	Person-to-Person
3R	e-waste reduce, recycle, and reuse strategy
RCI	Rwanda Cooperation Initiative
SIGFiP	Integrated Public Finance Management System
SIGPTS	Integrated Population Management and Secure ID System
SME	Small and Medium-sized Enterprise
SSA	Sub-Saharan Africa
STEM	Science, Technology, Engineering, and Mathematics
STEP	Systematic Tracking of Exchanges in Procurement
TA	Technical Assistance
ToR	Terms of Reference
TPM	Third Party Monitoring
TVET	Technical Vocational Education and Training
UN	United Nations
UNESCO	United Nations Educational, Scientific, and Cultural Organization
UNICEF	United Nations Children's Fund
USF	Universal Service Fund



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**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies)	Operation Name		
Chad	Chad Digital Transformation Project		
Operation ID	Financing Instrument	Environmental and Social Risk Classification	
P180000	Investment Project Financing (IPF)	Substantial	

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☒ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☒ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date
25-Sep-2024	30-Apr-2029
Bank/IFC Collaboration	Joint Level
Yes	Complementary or Interdependent project requiring active coordination

Proposed Development Objective(s)

To expand access to affordable and resilient broadband connectivity and improve government capacity to deliver public services digitally.

Components

Component Name	Cost (US\$)
Component 1. Climate-resilient broadband connectivity and digital inclusion	89,000,000.00



Component 2. Laying key foundations for digital government services	27,200,000.00
Component 3. Project management and knowledge transfer	6,000,000.00
Component 4. CERC	0.00

Organizations

Borrower: Republic of Chad

Implementing Agency: Ministry of Communications, Digital Economy, and Digitalization of Administration

PROJECT FINANCING DATA (US\$, Millions)**Maximizing Finance for Development**

Is this an MFD-Enabling Project (MFD-EP)? Yes

Is this project Private Capital Enabling (PCE)? Yes

SUMMARY

Total Operation Cost	122.20
Total Financing	122.20
of which IBRD/IDA	92.20
Financing Gap	0.00

DETAILS**World Bank Group Financing**

International Development Association (IDA)	92.20
IDA Grant	92.20

Non-World Bank Group Financing

Commercial Financing	30.00
Unguaranteed Commercial Financing	30.00

IDA Resources (US\$, Millions)



	Credit Amount	Grant Amount	SML Amount	Guarantee Amount	Total Amount
National Performance-Based Allocations (PBA)	0.00	92.20	0.00	0.00	92.20
Total	0.00	92.20	0.00	0.00	92.20

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2025	2026	2027	2028	2029
Annual	10.00	15.00	20.00	33.00	14.20
Cumulative	10.00	25.00	45.00	78.00	92.20

PRACTICE AREA(S)**Practice Area (Lead)**

Digital Development

Contributing Practice Areas

Governance; Education; Finance, Competitiveness and Innovation

CLIMATE**Climate Change and Disaster Screening**

Yes, it has been screened and the results are discussed in the Operation Document

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category	Rating
1. Political and Governance	● High
2. Macroeconomic	● High
3. Sector Strategies and Policies	● Substantial
4. Technical Design of Project or Program	● Substantial



5. Institutional Capacity for Implementation and Sustainability	● Substantial
6. Fiduciary	● Substantial
7. Environment and Social	● Substantial
8. Stakeholders	● Substantial
9. Other	● Substantial
10. Overall	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL

Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant
ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
ESS 8: Cultural Heritage	Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant



NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

LEGAL

Legal Covenants

Sections and Description

Schedule 2. Section I. A. 2: Not later than three (3) months after the Effective Date, the Recipient shall establish and maintain throughout Project implementation a strategic orientation committee at ministerial level: (a) with adequate resources and terms of reference satisfactory to the Association; (b) to be chaired by MCENDA and which shall include, inter alia, the ministers (or their representatives) of key ministries involved in Project implementation; and (c) which shall be responsible to oversee the general direction of the Project, providing strategic guidance and ensuring consistency with national development policies ("Ministerial Strategic Orientation Committee").

Schedule 2. Section I. A. 3: The Recipient shall maintain throughout Project implementation a high-level committee: (a) with adequate resources and terms of reference satisfactory to the Association; (b) chaired by the Secretary General of the Ministry of Finance, Budget, Economy and Planning and which includes representatives of the Recipient's key ministries involved Project implementation; and (c) responsible for providing overall guidance, strategic decision-making, approving the Annual Work Plan and Budget ("AWPB"), and overall oversight and coordination of Project implementation ("Steering Committee").

Schedule 2. Section I. A. 4: The Recipient shall maintain throughout Project implementation, a technical committee: (a) with adequate resources and with competent staff in adequate numbers and with terms of reference, qualifications, and experience satisfactory to the Association; (b) chaired by the technical Director-General of MCENDA, and composed of the Directors General of the key ministries involved in Project implementation; and (c) responsible for providing technical support to the Steering Committee and Project activities ("Technical Committee").

Schedule 2. Section I. A. 5: the Recipient, through the PIU, shall recruit, not later than three (3) months after the Effective Date: (a) a digital infrastructure technical lead; (b) a digital payments technical lead; (c) an e-government technical lead; and (d) an internal auditor, all with terms of reference, qualifications and experience acceptable to the Association.

ESCP: Adopt the SRA prior to Effective Date and thereafter implemented throughout Project implementation.

ESCP: Establish the grievance mechanism no later than ninety (90) days after the Effective Date, and thereafter maintain and operate the mechanism throughout Project implementation.

Conditions

Type	Citation	Description	Financing Source
Effectiveness	4.01	The Recipient has prepared and adopted a Project Implementation Manual in form and substance acceptable to the Association.	IBRD/IDA



Effectiveness	4.02	The Recipient, through the Project Implementation Unit (“PIU”), has recruited a Project coordinator, a financial management specialist, a social safeguards specialist, an environmental safeguards specialist, an accountant and a procurement specialist, all under terms of reference, and with qualifications and experience acceptable to the Association.	IBRD/IDA
Disbursement	Schedule 2. Section III.B.1 (b)	No withdrawal shall be made under Category (2) for Subsidies until the Recipient has: (i) prepared and adopted the Subsidies Manual, which forms part of the Project Implementation Manual; and (ii) has concluded at least one Subsidy Agreement with an existing Licensed Commercial Operator.	IBRD/IDA



I. STRATEGIC CONTEXT

A. Country Context

1. **Chad is a fragile, land-locked, low-income country with a population of 18.3 million, experiencing political instability.**¹ Agriculture and livestock employ 75 percent of the population, most of them engaged in subsistence activities. The oil sector dominates Chadian economy accounting for around 30 percent of Gross Domestic Product (GDP) and 70 percent of state revenue. Chad's GDP per capita was US\$722 in 2023, compared to an average of US\$757 in low-income countries and US\$1,623 in Sub-Saharan Africa.² Poverty, which had declined since 2003, increased in 2020 due to the COVID-19 pandemic and has remained high since, with over 40 percent of Chad's growing population living in poverty, predominantly in rural areas³. Chad ranks near the bottom of the United Nations' (UN) Human Development Index (187th out of 189 in 2021).

2. **Since early 2020, Chad has been subjected to multiple shocks.** The recent recession was mainly due to a temporary suspension of oil production and closing borders to contain the COVID-19 pandemic. In April 2021, a rebel incursion led to the death of President Deby. The presidential election to mark the return to constitutional order took place on May 6, 2024. Chad's constitutional council has confirmed Mahamat Idriss Deby as the winner of this election. Chad is the world's fifth largest receiving country of refugees in relative terms, with 1.1 million forcibly displaced persons in 2023.

3. **Chad is considered the world's second-most vulnerable country to the adverse effects of climate change (NDGAIN 2023).**⁴ The country is exposed to high risks of urban and river floods, droughts, water scarcity, extreme heat and wildfires.^{5,6} Mean temperature is projected to increase around mid-century by 3°C, along with projected increases in seasonal rainfall and the proportion of rainfall as heavy events will lead to flooding.⁷ These extreme climate events can adversely impact digital infrastructure and power supply. Accelerating the roll-out of mobile money and Digital Financial Services (DFS) can be leveraged to improve access to finance to improve climate resilience⁸.

4. **Chad's population is young and growing quickly, putting pressure on the country's resources.** Chad has the third youngest population in the world, with two-thirds of the population estimated to be younger than 25 years old. The fertility rate was an estimated 6.3 births in 2020, significantly higher than the SSA average. The population is projected to grow from 17 million in 2021 to 22 million by 2030.⁹ Population density is the highest in southern regions and in central regions near the borders with Cameroon, Central African Republic, Nigeria, and Sudan¹⁰. The weak capacity of the private sector makes it challenging to absorb the large cohort of young job seekers. Youth unemployment is a significant challenge,

¹ Unless otherwise noted, data in this paragraph are from the Macro Poverty Outlook, April 2024,

http://macropovertyoutlook.worldbank.org/mpo_files/mpo/mpo-sm24-tcd-scope.pdf

² Ibid.

³ INSEED, 2024, Ecosit 5

⁴ <https://gain.nd.edu/our-work/country-index/>. NDGAIN stands for Notre Dame Global Adaptation Initiative.

⁵ <https://thinkhazard.org/en/report/50>

⁶ Em-Dat The International Disaster Database

⁷ <https://climateknowledgeportal.worldbank.org/country/chad/climate-data-projections>

⁸ CCDR – Sahel 5 <https://www.worldbank.org/en/news/infographic/2022/09/19/g5-sahel-region-country-climate-and-development-report>

⁹ <https://futures.issafrica.org/geographic/guide.pdf?geography=TD&topic=current-path>

¹⁰ <https://reliefweb.int/map/chad/chad-population-density-2020>



as the number of young people entering the job market each year increased from 140,000 to nearly 210,000 between 2012 and 2020.¹¹

5. **Gender inequality in Chad is among the highest in the world**, and little progress has been made in the past decade. Literacy stands at 40.7 percent of young males compared to 22.4 percent of young females in 2021, and girls are lagging in primary, secondary, and tertiary enrollment and completion. Only 1 percent of women are employed in waged and salaried jobs compared to 11.4 percent of men. Women are underrepresented in public life and decision-making spaces, holding 14.9 percent of seats in national parliaments and 14.3 percent at ministerial level positions.¹²

6. **Diversification of the economy and infrastructure was recognized in Chad's latest National Development Plan (Plan National de Développement, PND) 2017-2021 as key to improving socio-economic outcomes.** The Ministry of Telecommunications and Digital Economy (*Ministère des Communications, de l'Economie Numérique, et de la Digitalization de l'Administration*, MCENDA), elaborated a new Strategic Plan for the Development of Digital and Post (*Plan Stratégique Développement du Numérique et des Postes*) 2020-2030 to “accelerate Chad’s digital transformation” and position the country as a “crossroads for ICT¹³ in Africa”. The plan aims at modernizing infrastructure, integrating digital technologies into growth sectors, support the development of financial services and financial inclusion, and creating jobs.

B. Sectoral and Institutional Context

7. **Chad's digital economy is nascent; many of the foundational building blocks needed to propel digital transformation at scale are underdeveloped or missing.**¹⁴ The present level of broadband access, which stands at 3 percent, is among the lowest in the Central African sub-region as well as in the Sahel. Weak digital adoption by consumers, businesses, and government means that digital technology is yet to have the desired transformational impact it could through contributions to growth, job creation, and expanded service delivery.

8. **Over the past two decades, the Government of Chad (GoC) has established several dedicated agencies and initiatives to advance the country's digital transformation**, including: (i) the Regulatory Authority for Electronic Communications and Post (*Autorité de Régulation des Communications Électroniques et des Postes*, ARCEP), established in 2014; (ii) the ICT Development Agency (*Agence de Développement des TIC*, ADETIC), established in 2014 to implement the national digital agenda;¹⁵ and (iii) the National Agency for Computer Security and Electronic Certification (*Agence Nationale de Sécurité Informatique et de Certification Electronique*, ANSICE), established in 2015 to coordinate the fight against cybercrime, ensure personal data protection, and secure electronic transactions.

9. **Since 2014, the GoC has taken measures to adopt a legal and regulatory framework that supports digital transformation, but existing laws would benefit from modernization or closer alignment with current good international practice**¹⁶. Implementation gaps exist due to the absence of implementing regulations and capacity constraints of key implementing agencies. Some provisions of national laws are outdated.¹⁷ There are still legal and

¹¹ Chad - Country Economic Memorandum : Boosting Growth and Reducing Vulnerability (English). Washington, D.C. : World Bank Group. <http://documents.worldbank.org/curated/en/099110723141052687/P1772530f4c3270c30ac45088b973c1a8e6>

¹² Chad Gender Landscape <https://liveprod.worldbank.org/en/economies/chad>

¹³ Information Communication Technology

¹⁴ World Bank, 2023. Chad Digital Country Diagnostic (P177120).

¹⁵ Also initiatives promoting ICTs, the design and delivery of digital services and platforms, training in digital skills in cooperation with the National ICT School (*Ecole Nationale Supérieure des TIC*, ENASTIC), and the provision of shared ICT services and e-government collaborative tools.

¹⁶ The Development Policy Financing series P163424-P168606, supported (i) the establishment of an open access wholesale Public Private Partnership (PPP) for the government network to Cameroon and Sudan, and (ii) transparency increase for institutions with the publication of financial statements by ADETIC and ARCEP.

¹⁷ Electronic communications (No. 013/PR/2014 & No 014/PR/2015), consumer protection (No. 005/PR/2015), personal data protection (No. 007/PR/2015), electronic transactions (No. 008/PR/2015), and cybersecurity and cybercrime (No. 009/PR/2015).



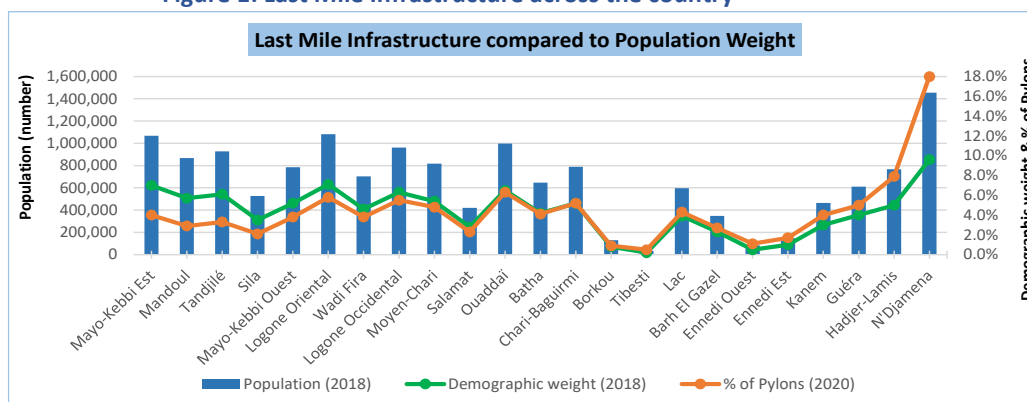
regulatory and institutional gaps in critical areas, such as roaming, number portability, and operation of the universal service fund (USF). Other gaps include clear rules on managing critical infrastructures to protect from cyber incidents, support interoperability, open data, and the security of electronic transactions.

10. **The GoC has initiated several structural projects in support of the country's digital infrastructure:** (i) an Internet Exchange Point (IXP) at ADETIC; (ii) the Electronic Communications Infrastructure Modernization Project, financed by Exim Bank China (US\$188 million)¹⁸, which aims to build 1,200 kilometers of optical fiber, a National Data Center (NDC), as well as 200 mobile telephone sites for the state-owned incumbent operator Sotel Tchad; and (iii) the Trans-Saharan Corridor Project (US\$30 million) funded by the European Union (EU) and managed by the African Development Bank (AfDB), which includes the construction of 550 kilometers of optical fiber between Massaget (Chad) and N'Guiguimi (Niger) under a Public Private Partnership (PPP). With the ongoing conflict in Sudan, the Eastern backbone is out of service, and the N'Djamena-Kousseri (Cameroon) link is the only operational international exit.

Gaps in broadband access

11. **Chad depends on its neighboring countries for international connectivity and faces last mile digital infrastructure access challenges,¹⁹ compounded by a low electrification rate.** Less than half of the population has access to broadband connectivity, and only 15.3 percent have broadband subscriptions (Global System for Mobile Communications Association [GSMA], 2022). In addition, 99.85 percent of these are mobile subscribers, and only 0.7 percent of women (versus 6.7 percent of men) report using the internet on a regular basis.²⁰ The two private operators, Moov Africa Chad and Airtel Chad, are the main providers of mobile broadband services since the cessation of activities of Salam, the mobile branch of Sotel Tchad. Chad lacks reliable access to electricity (11 percent compared to SSA average of 48 percent), which is a prerequisite to deploying digital services. Outside of urban areas, there is no grid electrification, leaving most rural areas with no electricity.

Figure 1: Last Mile Infrastructure across the country



Source: ARCEP Chad 2020 Markets Observatory Report and National Institute for Statistics, Economic, and Demographic Studies (Institute National de la statistique, des Etudes Economiques et Démographiques, INSEED) Chad 2018 Trend scenario

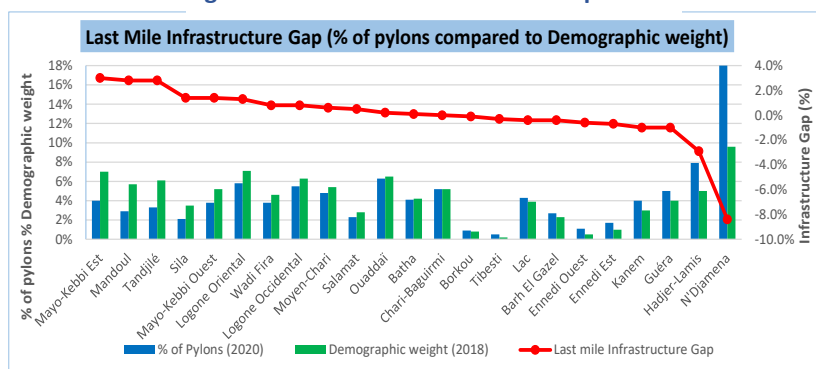
¹⁸ <https://presidence.td/synthese-du-mercredi-le-8-juillet-2020/3754/>

¹⁹ Chad is vast with hard-to-reach areas, landlocked (no direct access to submarine cable) and sparsely populated (13 inhabitants per km²). World Bank, 2022. World Development Indicators. Available at: <https://data.worldbank.org/indicator/EN.POP.DNST?locations=TD>

²⁰ The World Bank. Chad Digital Economy Diagnostic (2023).



Figure 2: Last Mile Infrastructure Gap



Source: ARCEP Chad 2020 Markets Observatory Report and INSEED Chad 2018 Trend scenario

12. **Last mile digital infrastructure is not spatially distributed in proportion to demographic weight of the regions.**²¹

The disparities in the availability of digital connectivity infrastructure seems to be at the border of Cameroon, Central Africa Republic, and Sudan (Figures 1 and 2 above). There is a need to understand the reasons of the market failure in these areas and assess the way public intervention can help expand broadband access to the population.

Unaffordable broadband data services

13. **Deployment and operation of broadband connectivity infrastructure is expensive in Chad, and broadband data services are unaffordable for average Chadians.** With a population density of 13 inhabitants/km²²², the cost of providing digital services is extremely high, and further exacerbated when factoring the cost of energy. Taxation specific to the ICT sector has increased since 2018 from 7 percent of the turnover of the sector to 9 percent in 2022. Consequently, the price for data services is among the highest in the Sahel and Central African regions (Table 1). The retail market, 99 percent dominated by the mobile segment, is a near duopoly between Moov Africa (with 52.6 percent of subscribers) and Airtel (with 47 percent). The fixed retail broadband market is essentially nonexistent. The wholesale broadband market is shared between the incumbent operator and Safitel²³, a wholesale infrastructure operator under a PPP, which operates the GoC's fiber assets on an open access basis.

Table 1: Affordability of 1 GB mobile data in the Sahel region and selected CEMAC countries

Country	Average Price of 2GB of Mobile Data as percentage of Gross National Income (GNI) per capita in 2022	Average Price of 2GB of Mobile Data in 2022	Monthly GNI per capita in 2022
Central African Republic	23.8%	US\$9.52	US\$40.00
Chad	22.5%	US\$12.93	US\$57.50
Burkina Faso	9.9%	US\$6.93	US\$70.00
Mali	9.7%	US\$6.87	US\$70.83
Niger	7.2%	US\$3.66	US\$50.83
Cameroon	3.7%	US\$5.12	US\$138.33

Source : 2022 Average price of 2GB (<https://www.cable.co.uk/mobiles/worldwide-data-pricing/>) and 2022 GNI per capita (<https://data.worldbank.org/indicator/NY.GNP.PCAP.CD>)

Digital Public Infrastructure and Services

²¹ <https://arcep.td/rapports>

²² <https://donnees.banquemondiale.org/indicateur/EN.POP.DNST?locations=TD>

²³ Following the challenging situation in Sudan, the fiber link interconnection has been interrupted and the former public joint-venture with Sudan Safitel has been entirely privatized with its capital owned by Tchadian private sector.



14. **The GoC has made progress in establishing some of the essential components for a digital government, but there is still work to be done.** ADETIC is making progress in interconnecting ministries, departments, and agencies (MDAs), but most government offices lack access to sustainable broadband internet, as well as the ability to exchange data through a common inter-governmental framework. Lack of leadership and of a whole-of-government approach for public sector modernization, and deficiencies in digital skills among public servants have all contributed to a limited number of digital public platforms. In the meantime, ministries are turning to private hosting solutions. Chad ranks at the bottom of the 2022 global e-government development index (189th out of 193 countries), well behind its peers such as other Sahelian countries (e.g., Burkina Faso 166th and Mali 168th) or CEMAC countries (e.g., Cameroon 141st or Republic of Congo 161st).²⁴

15. **The GoC has launched a few digitalization initiatives, mostly focused on Government-to-Government (G2G) services.** These aim primarily to digitalize government back-office systems or core government functions, such as public financial management and identity management.²⁵ These systems have been developed in a siloed approach, without common technical and security standards and guidelines, and through different funding sources (domestic and external aid). This limits interoperability, efficiency, and sustainability, and prevents them from effectively exchanging data and providing integrated services. For instance, Integrated Public Finance Management System (SIGFiP) was developed by the MFBEP IT unit with support from the Rwanda Cooperation Initiative (RCI) and is currently only operational at the budget and payroll levels.

Digital identification, e-procurement, and e-health government platforms

16. **Chad has prioritized the modernization of its identification (ID) and civil registration (CR) systems within its digitization agenda, but much is yet to be accomplished.** For countries like Chad with nascent digital economies, focusing on the building blocks of inclusive, trusted, and digitally enabled civil registration and basic identification is essential to move towards more advanced forms of digital ID that can be used to access online and remote services. Birth registration is 26 percent.²⁶ Ownership of national ID is 39 percent with a significant gender gap. Only 8 percent of the population has been issued the recently established Unique Identification Number (NNI). This can be attributed to several factors²⁷ impacting both the supply and demand of these services. Efforts are being made to increase coverage through improvements in accessibility of registration centers, but additional resources are needed to scale up. With support from UNICEF (with EU funding), the GoC is currently setting up birth registration kiosks in eleven health centers (in N'Djamena), equipped with digital tools. Chad's transition from predominantly paper-based and cumbersome CR processes to digitalization is accompanied by complex changes in institutional arrangements. Responsibilities are being transferred from the Ministry of Territorial Administration and Decentralization (*Ministère de l'Administration du territoire et de la Décentralisation*, MATD) to a relatively new National Secure ID Agency (*Agence Nationale Des Titres Sécurisés*, ANATS) who now plays a transversal role for the management of both digitalized CR and ID systems. This will require legal and technical support as well as a strong emphasis on data management practices.

17. **Despite some improvements in the legal and regulatory framework, Chad's procurement system requires significant enhancement.** Thresholds for prior review and approval remain some of the lowest in Africa. The regulations clearly define the integration of procurement into the budget cycle, but the practice differs, resulting in arrears and

²⁴ United Nations Department of Economic and Social Affairs, 2022. E-Government Survey 2022: Digital Government in the Decade of Action for Sustainable Development. Available at: <https://desapublications.un.org/sites/default/files/publications/2022-09/Web%20version%20E-Government%202022.pdf>

²⁵ Integrated Public Finance Management System - *Système Intégré de Gestion des Finances Publiques* (SIGFiP), eTax application, Automated System for Customs Data – ASYCUDA World and integrated identity management system, SIGPTS.

²⁶ UNICEF, Children under the age of five who have their birth registered.

²⁷ Predominantly paper-based processes, plus lack of awareness, of challenges in access to registration points, make it difficult to verify or replace documentation, and susceptible to damage, loss, or fraud. Most of professional CR officers (mayors, sous-prefects, or volunteers involved in civil registration are not trained professionals equipped to perform the mandate.

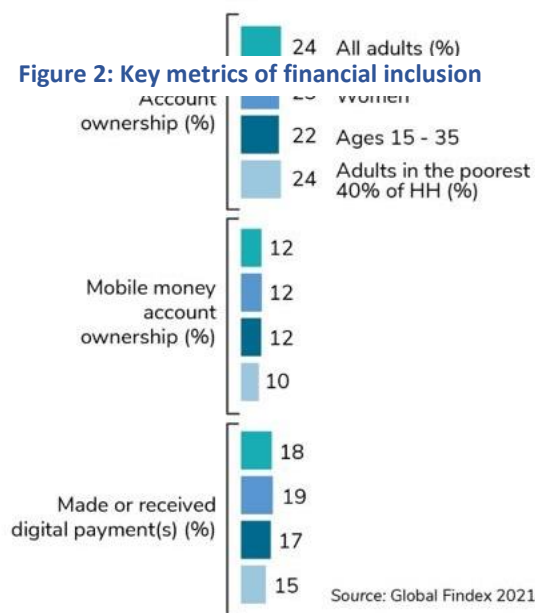


increased domestic debt. While the legislative and regulatory framework for public procurement and contract availability should be easily accessible, the service websites do not function effectively. Bid openings have not been public since 2020, restricted only to bidders. The procurement process reveals minimal information about participating parties, hindering transparency. In June 2023, the EU and the World Bank have both agreed to support the gradual design and implementation of an auxiliary information system for managing public procurement and interfacing it with the other public finance management applications.

18. **The COVID-19 pandemic crisis led to the emergence of Educational Technologies (EdTech) and digital health initiatives to ensure health delivery and education continuity, prevent service interruptions, and cope with social distancing.** Initial efforts have been made to digitize the learning format, with initiatives such as EduTchad²⁸ and Tchad-Educatic²⁹ launched in 2021. Other initiatives have involved using low-tech solutions, such as producing multimedia content through filmed courses in a studio. Similarly, the pandemic has highlighted the importance of digital health solutions, such as telemedicine and remote patient monitoring. As a result, there has been a surge in the development of digital health technologies and the GoC has the ambition to design and implement a harmonized and comprehensive National Telemedicine and Health Hotline Services program. These initiatives have the potential to transform the education and healthcare sectors, making them more accessible, efficient, and resilient in the face of future crises. However, such initiatives should be mutualized and also require the adoption of relevant digital skills, where the country lags behind.

Financial inclusion and digitization of payments

19. **The digitalization of financial services, including government payments, could bring efficiency and transparency while supporting innovation, competition, and financial inclusion.** The Global Findex 2021 finds that 24 percent of adults in Chad have a financial account in 2022, stagnant since 2017, compared to an average of 55.1 percent in SSA. While the share of adults with an account at a formal financial institution doubled since 2017 to 19 percent, it remains largely below the SSA average (39.7 percent in 2022). Only 12 percent of adults report having a mobile money account, and 6 percent effectively use a mobile money account two or more times per month.³⁰ In Chad, digital financial inclusion is not gender-neutral, as there are large gaps in access levels between men and women. The 2022 Findex data also reveals that 13 percent of women and 15 percent of men used a mobile phone or the internet to send money, while 12 percent of women and 13 percent of men received digital payments.³¹ In addition, only 4 percent of women saved money using a mobile money account (versus 5 percent of men).³² Gender gaps in the usage of digital financial services can be attributed to sociodemographic and cultural biases, limited access to credit, lower participation in training and initiatives in financial education in addition to a limited adaptation of digital financial services tailored to the needs of



²⁸ Developed in-house with the support of ADETIC, this was the first distance learning platform set up by the Ministry of National Education to be able to publish online digital educational resources produced to ensure pedagogical continuity during the COVID-19 period.

²⁹ Another distance learning platform, supported by the Embassy of France and with more features than EduTchad, but under a proprietary license with a private provider. It offers more than 4,000 courses in video, audio, and PDF formats, and partially accessible without internet connection.

³⁰ In 2021, there were only 176,798 active mobile money accounts. Only 3.6 percent of adults receive government payments in digital form.

³¹ The World Bank Group, The Global Findex Databased 2021.

³² The World Bank Group, The Global Findex Databased 2021.



female customers.³³ Chad is one of the few countries, and the only in SSA, where mobile money usage decreased between 2017 and 2021, from 15.2 percent to 12 percent, due to tax burden and regulatory measures following the illegal use of phone credit units for peer-to-peer (P2P) transfers. Several financial regulations limit the issuance and use of mobile money by microfinance institutions. Although some government services and social programs allow the use of mobile money (e.g., payment of community teachers and programs of cash transfer for displaced populations in some areas outside the main cities), a shared and integrated government digital payments infrastructure is not available. The GoC makes limited use of digital means of payment for inflows and outflows beyond a few use cases (e.g., payment of civil servants' salaries). In 2022, only 15 percent of adults received government payments³⁴ driven by payments of social safety nets programs. With respect to digital financial literacy, the 2022 Findex data continues to record gaps according to education and location, but less in terms of gender: 18.7 percent of women and 17.5 percent of male made or received a digital payment.

20. **Closing the gap requires complementing government payment digitalization with programs that build digital and financial literacy, along with establishing a strong consumer protection framework.** The GoC aims to accelerate adoption of digital payments and reduce over-reliance on cash. However, several constraints impede greater digitalization of transactions – notably Government-to-Person (G2P) and Person-to-Government (P2G). These include: (i) gaps in access to DFS and cash flow issues at points of service; (ii) a limited number of platforms and services conducive to digital payments; (iii) incomplete legal framework; and (iv) lack of interoperability among mobile wallet providers and other payment platforms. Weaknesses in the controls applied by electronic payments platforms need to be addressed, to prevent, detect, and suppress illicit trafficking and financial transactions. Chad has embarked since 2022 in enhancing its safeguard financial integrity, with a National Risk Assessment (NRA) completed in 2023.

Gaps in digital skills and literacy

21. **Gaps in digital skills and literacy pose a considerable obstacle to Chad's digital transformation.** No national policy or strategy dedicated to digital skills has been developed yet. The Technical Vocational Education and Training (TVET) system, which is fragmented and uncoordinated, has poor infrastructure and does not integrate digital skills. There is low tertiary educational attainment and a limited number of highly skilled professionals in the digital sector. Chad is particularly characterized by women's low enrollment in Science, Technology, Engineering, and Mathematics (STEM) fields and ICT degree programs.³⁵ It is reported that women's tertiary enrollment is as low as 1.5 percent (versus 5 percent for men)³⁶ while women only account for 4.8 percent of the researchers.³⁷ In 2019, less than 1 percent of women reported having any digital skills, while only 3.4 percent of men reported having at least one out of nine digital skills measured by the Multiple Indicator Cluster Survey.³⁸ Consequently, women participate in the labor market at a lower rate than men (64.4 versus 77 percent)³⁹, and are under-represented in the digital economy. The local leading STEM institutions are the University of N'Djamena, the Institute of Science and Technology of Abéché, the Polytechnic University of Mongo, and ENASTIC. ENASTIC conducts scientific and applied research and prepares its students for career pathways in local digital firms through an internship program. Finally, there are limited digital skills among public servants, which impedes the

³³ The World Bank Group, Chad Digital Economy Diagnostic (2023).

³⁴ Findex 2021.

³⁵ The World Bank Group, Chad Digital Economy Diagnostic (2023).

³⁶ The World Bank Group, Chad Digital Economy Diagnostic (2023).

³⁷ United Nations Educational, Scientific, and Cultural Organization (UNESCO), Women in Higher Education: Has the Female Advantage Put an End to Gender Inequalities? (2021).

³⁸ The World Bank Group, Chad Digital Economy Diagnostic (2023).

³⁹ The World Bank Group, Chad Digital Economy Diagnostic (2023).



adoption of digital government services. The GoC's limited technical and administrative capacity to harness digital technologies can be explained by the limited supply of digital skills.

C. Relevance to Higher Level Objectives

22. **The proposed project is aligned with Chad's Country Engagement Note (CEN) for FY23-24 (Report No. P500092),** discussed by the World Bank Board in November 2022, which supports the country's second Five-Year Development Plan. The project includes activities to help reach CEN Objective 2: Improve Governance and Transparency, and CEN Objective 3: Increase inclusive access to basic services and infrastructure. The project would also contribute to achieving Chad's Prevention and Resilience Allocation Action Plan, through support for activities to help achieve Strategic objective 2: Improve governance and strengthen political dialogue and Strategic objective 3: Increase access to basic social services and enhance local development strategies, especially in rural, border, and conflict-affected areas.

23. **The project contributes to several GoC and World Bank strategies, the African Union (AU) Digital Transformation Strategy, and IDA-20 Digital Commitments.** The sub-Axis 3.3 of the upcoming PND 2024-2028 on the development and access to quality infrastructure will serve as the high-level framework for this project. Access to affordable ICTs has been identified in Chad's 2030 Vision (*Vision 2030: Le Tchad que nous voulons*) as "an undeniable lever for development" which would "contribute significantly to the country's emergence." Project activities are aligned with the World Bank Western and Central Africa Regional Priorities for 2021–2025, in which broadband connectivity is seen as critical for economic transformation and creation of better jobs for more people. The project also supports the World Bank's Digital Economy for Africa (DE4A) initiative⁴⁰ underpinning the operationalization of the AU Digital Transformation Strategy⁴¹. It also contributes toward the World Bank's IDA-20 digital commitments related to Human Capital, Governance and Institutions, Jobs and Economic Transformation, and Gender and Development (see Annex 3). Activities will contribute also to the three pillars under the World Bank Group Gender Strategy (2024-2030): (i) end GBV and elevate human capital (by improving women's digital literacy and skills); and (ii) expand and enable economic opportunities (by creating job opportunities and enabling a wider access to services); and (iii) engage women as leaders (by involving women in decision-making processes).

24. **The project contributes directly to the Accelerating Digitalization Global Challenge Program (GCP).** Through its two main components, the proposed project will implement the GCP three Strategic Focus Areas (see paragraphs 29 and 34). The project is also aligned with the priorities emphasized in the World Bank's World Development Report 'Data for Better Lives'. It will enhance data governance and security in Chad as well as improve the capacity of the GoC to better manage and utilize data through strengthening data governance and management frameworks, capacity building and investments in data exchange and security. The project will also strengthen data safeguards that build trust in the digital economy.

25. **The Project will support Chad in achieving its Nationally Determined Contribution and is consistent with country's strategies on climate change.** In the latest 2021 Nationally Determined Contribution update, Chad has committed to conditional emissions reduction target of 19.3 percent by 2030 compared to the business-as-usual scenario. Chad's Nationally Determined Contribution has prioritized energy efficiency and renewable energy and has outlined infrastructure as one of the key sectors for adaptation to which the project will contribute towards through its support for low-carbon and climate-resilient deployment of digital connectivity infrastructure (Component 1). Chad's National Adaptation Plan cites "management of risks and extreme weather events" and its CCDR⁴² prioritizes mainstreaming the

⁴⁰ <https://www.worldbank.org/en/programs/all-africa-digital-transformation>

⁴¹ which aims to ensure that every African individual, business, and government is digitally enabled by 2030, and is building on the Digital Economy Diagnostic (P177120)

⁴² World Bank Country Climate and Development Report (CCDR) for Chad was part of the Sahel G5 CCDR's analysis (<https://openknowledge.worldbank.org/server/api/core/bitstreams/197f07e0-d3ec-48bc-9d52-38d551d5d312/content>)



use of early warning systems and hydro-meteorological information which the project will enable by expanding digital access. Additionally, by providing access to digital connectivity and henceforth digitally enabled services such as digital ID and emergency communication response facilities, the project will strengthen climate resilience of the communities in climate hotspots.

II. PROJECT DESCRIPTION

A. Project Development Objective

PDO Statement

26. The PDO of the proposed operation is to expand access to affordable and resilient⁴³ broadband connectivity and improve government capacity to deliver public services digitally.

PDO Level Indicators

27. The achievement of the PDO will be measured by the results indicators below.

Expand access to affordable and resilient broadband connectivity.

- (i) People using broadband internet (new use) (Number; of which women, of which youth)
- (ii) Basket prices of broadband services (Percentage)

Improve government capacity to deliver public services digitally.

- (iii) People using digitally enabled services (new services) (Number; of which women, of which youth)
- (iv) People who made or received a digital payment (Percentage)

B. Project Components

28. **The project supports Pillars 1, 2 and 4 of the World Bank Group's Strategy for Fragile, Conflict and Violence (FCV) Countries 2020-2025, which recognizes that digital infrastructure can be used to tackle drivers of fragility and that digital technologies have great potential to support transitions from fragility to stability and help rebuild state resilience. With only two main components and simplified institutional arrangements, the proposed project design is adapted to the FCV and limited capacity context of Chad.** It also features private capital mobilization, climate adaptation and mitigation measures, and an emphasize on gender aspects, to implement the Accelerating Digitalization GCP and enable digital transformation in Chad. Table 2 summarizes the component structure.

⁴³Affordability refers to the ability of individuals and businesses to access and use broadband services without facing financial constraints. It encompasses the cost of both the hardware necessary to access the network (e.g., personal computers, smartphones) and the ongoing costs of broadband access plans. The affordability of broadband services is a significant barrier to achieving universal access, as it directly impacts the ability of individuals and businesses to utilize digital tools and services. For example, in the context of Chad, efforts to improve affordability focus on lowering prices and increasing the value derived from broadband connectivity, such as through productive uses of digital tools and services like e-commerce and e-learning.

Resiliency, on the other hand, refers to the ability of digital infrastructure and connectivity to withstand and recover from natural hazards, disruptions, and crises. This includes ensuring that infrastructure can function and meet users' needs during and after natural disasters, civil unrest, disease outbreaks, and climate-related disasters. Resilient connectivity supports citizens and emergency responders during crises, ensures business continuity, and allows for swift responses to trends and challenges, thereby increasing overall resilience. For instance, in the context of Chad, the extension of digital networks under an MFD approach aims to fill network coverage gaps in digital infrastructure, particularly in rural areas, and enhance network resilience to support citizens and emergency responders during crises.



Table 2. Project Components and Proposed Costs

Components	Financing (US\$ million)		
	IDA	Commer- cial	Total
Component 1. Climate-resilient broadband connectivity and digital inclusion	59.0	30.0	89.0
1.1. Strategic, policy, legal, regulatory, and institutional frameworks for climate-resilient broadband access	2.0		2.0
1.2. Rural broadband access	41.0	27.0	68.0
1.3. Urban and peri-urban broadband access	11.0	3.0	14.0
1.4. Training and digital inclusion	5.0		5.0
Component 2. Laying key foundations for digital government services	27.2		27.2
2.1. Strategic, policy, legal, regulatory, and institutional frameworks for trusted digital services	1.2		1.2
2.2 Technical foundations of e-government, trust services, and ICT capabilities of the public sector	7.0		7.0
2.3. Digitization of payments	10.0		10.0
2.4. Digitalization of selected services	9.0		9.0
Component 3. Project management and knowledge transfer	6.0		6.0
Component 4. CERC	0	0	0
TOTAL	92.2	30.0	122.2

Component 1: Climate-resilient broadband connectivity and digital inclusion⁴⁴ (US\$89 million equivalent, of which US\$59 million equivalent from IDA, and US\$30 million equivalent from the private sector)

29. This component will aim at closing the digital access gap by bringing affordable broadband services to rural communities and vulnerable populations, bringing public services (Component 2) and enhancing the presence of state in FCV areas to strengthening resilience by building digital capabilities and services for conflict zones (addressing drivers of fragility in East and North of Chad). Digital infrastructure will be subject to quality standards for climate resilient and low-carbon digital connectivity. This component will contribute to the GCP Strategic Focus 1 on Affordable Broadband. In line with Maximizing Finance for Development (MFD) principles and spatial targeting approaches, synergies with World Bank-funded ongoing and pipeline projects will be leveraged, especially in the Energy (Chad Energy Access Scale Up Project – P174495) and Transport (Cameroon-Chad Transport Corridor - P167798) sectors as well as Health and Education sectors. The design of digital connectivity infrastructure will consider projected increase in floods, rise in temperatures, and other extreme climate events identified in the climate risk assessments (to be conducted by the operator and built into the tender) and adequate adaptation measures will be embedded accordingly to climate-proof the digital connectivity infrastructure investments. The telecom network construction under this component will follow best-available energy efficiency standards in ICT energy efficiency which include: (i) requirements for renewable energy solutions (e.g., solar or wind power and battery storage in excess capacity for increased energy access) when off-grid to power the infrastructure, identification of opportunities for co-deployment of low carbon energy solutions.

30. All connectivity infrastructure deployment and procurement under Component 1 will be in accordance with energy efficiency and energy performance best international practices and standards such as Green ICT Standards⁴⁵ and/or recommendations from the European Commission's Joint Research Centre (JRC) report Best Environmental

⁴⁴ In 2024 alone, Chad has experienced several internet outages, overall country's digital resilience is ranked very low: Pulse Internet Resilience Index, ranks Chad as 2nd last for all African countries (Figure 2). <https://pulse.internetsociety.org/blog/chads-recent-outages-highlight-resiliency-gaps>

⁴⁵ ITU-T, Green ICT Standards and Supplements (<https://www.itu.int/net/ITU-T/lists/standards.aspx?Group=5&Domain=28>), IEEE 802.3az Energy Efficient Ethernet standard, ISO 14001 v2015: Environmental Management Standard, ISO 50001 v2011: Energy Management Standard, ISO/TR 14062 v2002.



Management Practice in the Telecommunications and ICT Services Sector⁴⁶ to the extent technically and financially feasible. The existing connectivity prevalent in Chad is a mix of energy intensive copper cables and microwave technology. Under this project, all cable deployment will be fiber optic cables that are the best available technology in Chad⁴⁷ and also substantially will be more-energy-efficient than microwave and traditional copper-based networks, which are still used in the country, triggering significant energy savings of approximately 40-60 percent⁴⁸. Extending fiber coverage (under Subcomponents 1.2 and 1.3) in Chad will help to reduce the size of the microwave backhauling network for backbone connectivity and last mile and will result in a significant reduction in greenhouse gas (GHG) emissions associated with the data traffic, as optical fiber is two to three times more efficient than microwaves for backhauling, allowing power savings from 54 to 94 percent on the backhaul deployment in a 4G scenario; and all telecommunications network equipment to be procured under this component will have energy efficiency levels that will surpass country's current energy efficiency levels and equipment procured for similar expansion projects⁴⁹. These will follow international practice guidelines for energy efficiency⁵⁰. For connectivity deployment and procurement, preparation of specifications and tenders (design, equipment, and operational modes) will include requirements for energy efficient hardware/software (such as network standby modes, power feeding solutions, and others) and requirements to power infrastructure using renewable energy to the extent possible.

Sub-component 1.1. Strategic, policy, legal, regulatory, and institutional frameworks for resilient and affordable broadband access (US\$2 million equivalent)

31. This sub-component will aim to strengthen broadband market competition and stimulate private sector participation in closing the digital access gap, in line with MFD principles.

(a) **Support to MCENDA and ARCEP to strengthen telecoms/broadband regulation** including : (i) gap analysis of the legal and regulatory framework to make it conducive to digital development; (ii) improving the observatory of broadband infrastructures and services; (iii) broadband wholesale and retail market regulation, spectrum management; (iv) liberalization of Unstructured Supplementary Service Data to promote the development of the fintech ecosystem; (v) strengthening of the sharing of infrastructure under a “dig once” approach that will significantly reduce GHG emissions⁵¹, including use of electricity poles to deploy fiber optic; (vi) reforms to improve the governance and management of international connectivity; (vii) set up of a certification laboratory; and (viii) purchase of equipment

⁴⁶ Canfora, P., P. Gaudillat, I. Antonopoulos, and M. Dri., 2020. “Best Environmental Management Practice in the Telecommunications and ICT Services Sector.” EUR 30365 EN, Publications Office of the European Union, Luxembourg, 2020, ISBN 978-92-76-21574-5, doi:10.2760/354984, JRC121781

⁴⁷ Without the World Bank's intervention and in a non-project scenario, the use of energy-intensive microwave (instead of the modern energy efficient fiber optic cables) for broadband connectivity would be far more extensive in Chad.

⁴⁸ Research has shown that copper cables consume more energy than fiber optic cables; copper networks consume about 3.5 W at full 100 m reach capability while fiber networks may use less than 1 W to transmit the 10 GbE signal over 300 m (<https://www.cablinginstall.com/cable/fiber/article/16465844/how-fiber-can-help-make-your-network-greener>). Fiber is also 70–80 percent more reliable than copper, which results in approximately 60 percent fewer maintenance callouts and energy savings between 40 percent and 60 percent; see Godlovitch, Ilsa, Peter Kroon, Sonia Strube Martins, and Fabian Eltges. 2019. “Copper Switch-off, A European Benchmark.” FTTH Council Europe Conference, Amsterdam, page 9

⁴⁹ There are no mandatory minimum energy performance standards and energy labeling system in Chad for network equipment and connectivity deployment (information obtained from ADERM *Agence pour le développement de l'électricité rurale et la maîtrise de l'énergie*)

⁵⁰ Such as The European Commission's JRC report Best Environmental Management Practice in the Telecommunications and ICT Services Sector

⁵¹ Evidence suggests that increase in number of operators per tower, for instance, leads to energy savings between 37-53 percent. Source: Helios Towers. 2021. [Sustainable Business Impact Report](#). Infrastructure sharing can lead to large energy savings. The baseline carbon emissions for 4G sites using a wireless backhaul is estimated to be 5.2 kt for 30 gigabytes (GB) per month per user (Figure 6), but the amount can be reduced by 37 percent when using an active sharing approach. The downstream impact is bigger when opting for fiber rollout as opposed to microwave or other. Much of the cost of fiber rollout relates to trenching, which can be reduced if there are existing ducts or poles. Source: World Bank. 2023. [Green digital Transformation](#)



for Quality-of-Service and tariff monitoring as well as spectrum management and control, including training for ARCEP staff.

- (b) **Support to MCENDA, ADETIC and ARCEP in developing and operationalizing a universal broadband access strategy to achieve universal access by 2030** through TA for (i) developing a broadband policy with a clear roadmap to expand rural broadband coverage under a Private Capital Enabling/Mobilization (PCE/PCM) and climate-resilient approach (i.e. ensuring continuity of service during climate events and natural disasters) and taking into account security and electricity constraints; (ii) an assessment of barriers to adoption of digital services, such as basic digital literacy, and gender-specific user constraints; and (iii) a strengthening of ARCEP's institutional framework to improve the governance of the USF funds, improve its transparency and efficiency for faster disbursements for rural connectivity projects.
- (c) **Analysis of climate change impacts on the digital sector and identification of the role of digital technologies (telecom and data infrastructure and services) to strengthen resilience to climate risks** such as floods, drought, water scarcity challenges etc., exploring means (legislation, regulations or guidelines) for introducing climate-informed sectoral regulation to address Chad's climate-specific vulnerabilities and embedding the requirements for digital network resilience and recovery recommended by the International Telecommunication Union (ITU)^{52, 53} into the telecom regulatory framework.
- (d) **Support for the development of an e-waste management policy framework**, based on quantitative and qualitative baseline surveys to assess the carbon footprint of e-waste in Chad and following a circular economy approach. The framework will also include the development of an e-waste reduce, recycle, and reuse (3R) strategy to address the full life cycle of electronic devices and equipment with the aim to reduce greenhouse gas emissions. The project would thereafter support the implementation of the 3R strategy to ensure the collection and disposal, recycle and reuse of e-waste at the community-level through partnership with non-governmental organizations (NGOs) and the private sector.⁵⁴

Sub-component 1.2. Rural broadband access (US\$68 million equivalent, of which US\$41 million equivalent from IDA and US\$27 million equivalent from the private sector)

32. This sub-component aims to address market failures preventing rural areas from having access to broadband internet, using catalytic public financing (reverse auctions⁵⁵). It will leverage private sector investments to expand broadband infrastructure⁵⁶ by deploying 500 solar-powered mobile sites⁵⁷ covering each at least 3,000 people. Proposed activities include:

⁵² ITU, 2014. Requirements for Network Resilience and Recovery. <https://www.itu.int/en/ITU-T/focusgroups/dnrrr/Documents/fg-dnrrr-tech-rep-2014-6-NRR-requirement.pdf>.

⁵³ Green ICT standards: <https://www.itu.int/net/ITU-T/lists/standards.aspx?Group=5&Domain=28>

⁵⁴ See example in Ghana: <https://www.scidev.net/global/features/upcycling-e-waste-trash-into-innovative-treasure/>

⁵⁵ The competitive selection is done based on a reverse auction, wherein the operator or the consortium selected to cover the underserved areas is the one requesting the minimum subsidy. This approach has been successfully supported by the World Bank in Côte d'Ivoire (E-Agriculture Project, P160418), Benin (Digital Rural Transformation Project, P162599), and Niger (Smart Villages for Rural Growth and Digital Inclusion, P167543).

⁵⁶ Based on site-specific climate risk assessment for mobile sites, climate adaptation measures will be included to climate-proof the infrastructure such as active and transmission equipment. Climate risks will be considered in the site-selection and construction of mobile towers and measures will be taken (e.g., constructing towers in areas that are at low-risk or embedding elevation/strengthening the base of towers to withstand floods). The infrastructure deployed and the equipment installed/procured will be energy efficient (as described under component 1).

⁵⁷ The operators will be required to equip new pylons with renewable sources such as solar systems for batteries and power-supplies.



- (a) **A TA package to inform investments to expand the geographic coverage of broadband connectivity for selected underserved/uncovered villages (US\$1 million equivalent).** In line with international best practices and the cascade approach, activities to be financed are (i) identify areas where there is a market failure in the provision of broadband services; (ii) target locations that best support the achievement of the PDO by striking a balance between locations particularly exposed to high climate risks (such as flooding and drought) with the maximum economic impact and areas with the poorest and most vulnerable populations – taking into account the security situation as well as electricity densification plans (including increasing the percentage of solar in the mix and allowing more people to charge their device); (iii) estimate the minimum public funding necessary (reverse auction process) to incentivize private operators to cover the targeted areas; (iv) provide support to the Government in bids assessment and contract negotiation related to the call for tender to deploy and operate broadband connectivity infrastructure in rural areas; and (v) supervise the construction works to ensure its compliance with all technical and safeguards requirements specified in the bidding documentation.
- (b) **Subsidies for the deployment of broadband last mile connectivity in lagging rural areas (US\$40 million equivalent), and recruitment of an independent firm responsible for the supervision of the construction process and that will ensure its compliance with all the technical and safeguards requirements,** including requirements for climate resilience, energy efficiency and use of solar power, specified in the bidding documentation. The deployment will aim to follow low-carbon, and climate-resilient approach (see details in para 29) and will have energy efficiency levels that will surpass country's current energy efficiency levels and for similar expansion projects.

Sub-component 1.3. Urban and peri-urban broadband access (US\$14 million equivalent, of which US\$11 million equivalent from IDA and US\$3 million equivalent from the private sector)

33. **This sub-component will provide last-mile connectivity access to selected public institutions, including MDAs, local government offices, healthcare facilities, and higher and basic education institutions, in N'Djamena, in secondary cities, in remote and fragile areas.** By bringing broadband connectivity to FCV areas, the project will contribute to improving the presence of state in the East and Northern provinces. The approach being applied for last-mile connectivity⁵⁸ under the proposed project is demand aggregation and prepurchase of capacity for the Government.⁵⁹ This approach ensures value for money, through competitively tendered contracts with the private sector that guarantee medium- to long-term supply agreements with the Government, which is enabling MFD. Activities under this sub-component include:

- (a) In line with international best practices and the cascade approach (US\$1 million equivalent), TA to: (i) determine the locations and the target number of public institutions based on a feasibility and mapping exercise, considering security and energy-access. The mapping exercise will be used to analyze and identify areas exposed to high-climate risks when designing and prioritizing network coverage to these unconnected climate hot-spots. The list of public institutions would reflect priority sites identified in close collaboration with various MDAs, as well as other ongoing World Bank-financed- projects to maximize synergies and development impacts and will also include higher education institutions for the purpose of connecting to a national research and education network (NREN); (ii) assess the need for deployment of Metropolitan Area Network (MAN) for government connectivity to support MDAs' digitalization efforts in urban areas and propose an approach to leverage private sector investment taking into account the security and

⁵⁸ The infrastructure deployed and the equipment installed/procured will be climate-resilient and energy efficient following best practices on energy efficiency (as described under component 1 description).

⁵⁹ Indefeasible Rights of Use (IRUs) are commercial capacity supply agreements that require the private sector to guarantee long-term capacity over a fixed route (e.g., fiber networks). The Government will provide bidders with the capacity and costs needed for its operations (study funded in Subcomponent 1.1) and will request the private operator to propose the most competitive cost for providing the service. The effective IRU cost will set a floor on the amount that investors will have to recover from end-user customers to break even.



energy-access; and (iii) prepare the bidding documents and support to the Government in bids assessment and contract negotiation.

- (b) **Prepurchase of broadband connectivity for government, under indefeasible right of use (IRU)⁶⁰ and NREN membership (US\$10 million equivalent).** It will also include climate resilient⁶¹ and energy efficient equipment⁶² for indoor connectivity (Wi-Fi routers, LAN, etc.) and energy access.

Sub-component 1.4. Training and digital inclusion (US\$5 million equivalent)

34. **This sub-component will focus on enhancing basic and intermediate digital skills of the population to strengthen internet adoption for productive purposes, including for using digital services.** They would build on existing initiatives inaugurated by ADETIC. Particular attention will be given to ensuring that the most vulnerable communities, including in remote rural areas, areas exposed to high climate risks such as flooding, women, youth, Internally Displaced Persons (IDPs) and refugees, and persons with disabilities (PwD), are prioritized. Proposed activities under this sub-component include:

- (a) **The development of climate-oriented training programs for young trainers to enhance basic and intermediate digital skills for internet adoption (US\$2 million equivalent).** They will target areas covered by infrastructure investments under Sub-components 1.2 and 1.3. Trainings would first be tested as pilots in selected communities, adopting an iterative approach and customized to the local context of Chad, before being expanded under the project. The trainings include: (i) digital literacy, financial literacy, cybersecurity, data protection, and consumer protection trainings; (ii) digital technologies that can be leveraged for climate adaptation and mitigation (e.g. early warning systems, digital cash transfers in the event of shocks, solar-powered devices, energy efficient equipment, etc.); and (iii) targeted outreach and initiatives for women and girls, including dedicated digital ambassador programs to teach rural communities how to use digital tools tailored to local needs. Within training communities, the project will include content and approaches designed to close observed gender gaps in access to digital devices and digital skills. This may include men's engagement programming and couples' interventions designed to ease restrictive social norms and promote women's uptake of the training programs and access and use of digital devices: recent impact evaluations have demonstrated the potential for light-touch interventions to yield large impacts.⁶³ Adaptation may also include tailoring training content, delivery modalities, and delivery timing to account for gender differences in functional literacy and recognition of women's household responsibilities. To boost demand for digital and STEM skills, the project will consider building on promising evidence on interventions to disseminate information on the returns to these skills.⁶⁴

⁶⁰ Any operations and maintenance costs that go beyond the closing date of the project, as well as additional bandwidth purchase after the closing date, would fall under the responsibility of the Government.

⁶¹ Adequate adaptation measures will be embedded such as weather-proofing the switches, sockets and appliances in the network, and ensuring continuity and availability of connectivity in strategic entities in case of possible disruption from extreme climate events.

⁶² Since there is no energy efficiency certification available for telecom equipment, all electronic fittings (equipment) and telecommunications network equipment to be procured under this component will have energy efficiency levels that will surpass country's current energy efficiency levels and equipment procured for similar expansion projects, using the best available technology (similar to description under component 1) Any ICT equipment used for LANs to connect government offices and training institutions will use energy-efficient technologies to support energy conservation and reduce GHG emissions: equipment with automatic switch-off mechanisms or that conserve power when not in use

⁶³ A recent impact evaluation of a couples training and planning intervention targeting the rubber value chain in Cote d'Ivoire yielded significant agricultural productivity increases while a norms intervention disseminating information on community support for women's land ownership boosted land registration in women's names. <https://www.sciencedirect.com/science/article/pii/S016726812100158X>.

⁶⁴ A recent impact evaluation showed that disseminating information on the returns to male-dominated sectors can increase the share of women seeking employment in these sectors: <https://www.journals.uchicago.edu/doi/abs/10.1086/726710>.



- (b) **Support for the set-up of additional ‘digital user centers’** (*US\$3 million equivalent*), which may be fixed (rented building or repurposed container will be refurbished⁶⁵ with training room, workstations, solar energy⁶⁶ where grid is not available or for backup, etc.) or mobile, to provide rural communities access to vital services, relevant local content including public services, digital literacy training and access to free or low-cost internet. The project will finance rehabilitation, equipment, and operational costs. The mobile digital user centers will be deployed in priority in fragile areas (e.g., Eastern and Northern parts of the country), addressing some drivers of fragility by improving (digital) skills and bringing public services to the most vulnerable populations⁶⁷. Hardware and IT equipment (such as computers, printers, scanners, projectors) and other electrical equipment (air conditioning units and lighting) will be energy efficient certified at levels that surpass the country’s existing energy efficiency levels for similar equipment (in line with internationally recognized certifications such as Energy Star)⁶⁸. In rural communities where there is limited grid connection, solar energy systems will be utilized. A site-specific climate risk assessment will be conducted to determine climate risks and accordingly incorporate adaptation measures in infrastructure (user centers) to make it resilient to adverse weather events and for business continuity in the face of extreme climate events. Specifically, the renovation design will involve using climate-proof, heat-resistant, and reinforced materials, and improved drainage.

Component 2: Laying Key foundations for digital government services (*US\$27.2 million equivalent*)

35. **This component will provide support to build GoC’s digital platforms, strengthen institutional capacity for digital public service delivery to the Chadian population, enhance data security and personal data protection capabilities, digitalize prioritized services and systems, and enhance the government’s operational efficiency.** As such, it is directly contributing to the implementation of the GCP Strategic Focus Areas 2 and 3. Priority digital services identified by the government for digitalization include digital payments – with a view to improving access for women and marginalized or vulnerable populations, including persons with disabilities and people in fragile areas or of high climate risk.

Sub-component 2.1. Strategic, policy, legal, regulatory, and institutional frameworks for trusted digital services (US\$1.2 million equivalent)

36. **This sub-component aims to strengthen the legal, regulatory, institutional, and strategic frameworks to support safe and trusted delivery of and access to private and public digital services.** It includes:

- (a) **Development of strategies, norms, and processes for government services digitalization through a long-term TA** (*US\$0.7 million equivalent*) including the development of a whole-of-government approach for: (i) standards and guidelines for public service digitalization, user-centric digital services, and data governance, exchange, management and security; (ii) an e-/digital government strategy offering an overarching architecture of the digital transformation integrating the climate risks, and including multi-channel access (single portal, mobile, and kiosks or one-stop shops), e-ID, digital payment, interoperability, and infrastructure (e.g. data center), broadband, and cybersecurity; (iii) institutional capacity building for strengthening digital governance of public sector; and (iv) change management activities, workshops, training and communication activities.
- (b) **Strengthening the legal and regulatory framework for key areas of the digital economy and to support online service delivery** (*US\$0.5 million equivalent*), including review of existing laws and regulations adopted between 2014 and

⁶⁵ Mitigation and adaptation measures will be integrated: Retrofits/repairs of existing buildings/containers are planned to ensure they integrate energy efficiency and low-carbon measures. Equipment financed for user centers will be certified energy efficient.

⁶⁶ The Project will identify renewable energy solutions (e.g., solar or other renewables and battery storage) to power the Digital user centers.

⁶⁷ Like the 150 digital centers being deployed in rural areas in Niger as part of the Smart Villages for Rural Growth and Digital Inclusion Project (P167543) or the 23 Digitrucks deployed as part of the Niger Learning Improvement for Results in Education Project (P168779).

⁶⁸ There are no national energy efficiency performance standards or energy labeling systems in Chad (see details under component 1)



2019 (data protection, e-transactions, cybersecurity and cybercrime, and the legal framework regulating CR and the ID system⁶⁹) to ensure they are fit for purpose. The TA will also support new laws and implementing regulations in areas that are necessary to the effective implementation of other activities, (e.g., access to information and open data laws, a legal framework for e-government to support data portability and interoperability, and implementing regulations for the legal framework for e-transactions, including to enable electronic certifications and signature (i.e., use of Public Key Infrastructure [PKI]), digital payments and ensure consumer protection. This TA will consist of assessing existing gaps, supporting the drafting of laws and regulations aligned with international best practice and adapted to the Chadian context, and support for public consultation to ensure a transparent and accountable process and address any unintended gaps. Finally, the project will finance regulatory capacity-building efforts in selected ministries to support the enforcement of this legal and regulatory framework and sensitization campaigns with target groups from the public and private sectors.

Sub-component 2.2. Enhancing foundations of digital government (US\$7 million equivalent)

37. This sub-component aims at strengthening building blocks for public service digitalization based on government-wide standards (whole-of-government approach). Financing will be provided for:

- (a) **Improving the security and efficiency of government (internal and external) communications and data exchange within the public administration (US\$2 million equivalent)**, through: (i) TA and investments for licensing arrangements, migration and operational support plans for secure government email and information collaboration; (ii) the establishment of a directory of public services, inventory of registries, feasibility study for an online one-stop-shop public service portal; and (iii) the development and implementation of information services for users of administrative services, including a mapping and assessment of vulnerable/refugee groups' demand on public services.
- (b) **Developing national data hosting capabilities,⁷⁰ facilitating data migration, and establishing robust data disaster recovery strategies⁷¹ (US\$2 million equivalent)** through: (i) TA for the development and implementation of national data hosting (including cloud), data migration, data disaster recovery strategies that will also include support to systems integration that is expected to generate energy efficiency gains; and (ii) the procurement of energy efficient solar-powered IT equipment for data server rooms⁷² and private cloud hosting solutions with backup solutions for select MDAs. On-site data hosting environment will be strengthened by integrating physical protection measures, energy conservation measures, data security, climate adaptation measures, in accordance with international best practices on energy⁷³.
- (c) **Development of government capabilities to mitigate cybersecurity risks, manage data protection and develop electronic certification (US\$1.5 million equivalent)**, through: (i) TA (*Assistance à Maitrise d'Ouvrage*, AMOA) to support the implementation of the national cybersecurity strategy, of the roadmap for critical infrastructure

⁶⁹ Adopting a whole-of-government approach to government digital transformation and a holistic approach to (digital) identification, strengthening interoperability while promoting secure, privacy-conscious data sharing will be essential to build a financially sustainable ID ecosystem.

⁷⁰ Climate vulnerability of current data hosting arrangements in the country will also be assessed to identify adequate adaptation measures to strengthen the data hosting environment (e.g., by designing a redundant data hosting system to increase resilience during extreme climate events).

⁷¹ This includes automated backup and disaster recovery to avoid data loss (e.g., hosting systems across multiple sites/cloud.) due to damage and service interruption resulting from extreme climate events.

⁷² The purchase of the energy-efficient equipment for data centers and cloud solutions will follow international best practices on energy efficiency such as Energy star certified or international standards on green procurement (e.g., Development of the EU Green Public Procurement (GPP) Criteria for Data Centres, Server Rooms and Cloud Services). A site-specific assessment for server rooms and other equipment will be conducted to assess risks (floods and intense rainfall) to inform the placement of equipment and other mitigation measures (e.g., elevating the height of servers).

⁷³ For example, the EU Code of Conduct on Data Centre Energy Efficiency or ITU recommendations on green data centers.



protection (CIP) program, and support project cybersecurity investments; (ii) independent data protection, information security and cyber vulnerability audits for selected public institutions; (iii) TA/AMOA to develop a long-term operational plan for implementing digital certificate management to enable seamless implementation of digital certification projects or PKI operation center to support the establishment of the root Certificate Authority system; (iv) development of a Data Protection Unit five-year strategic plan; (v) acquisition and installation of software and critical energy efficient IT equipment⁷⁴ for the national Computer Emergency Response Team (CERT), Security Operations Center, data protection and digital certification system or PKI; and (vi) provision of trainings, awareness and communication services on cybersecurity, online safety including gender-based violence and cyber harassment, data protection, and digital trust services.⁷⁵

- (d) **Enhancing the digital skills within the public administration workforce and infrastructure of Higher Education Institutions (HEIs) for facilitating the adoption and implementation of digital government programs** (*US\$1.5 million equivalent*), through: (i) a comprehensive evaluation of ENASTIC and relevant HEIs, which will inform Project investments in their infrastructure and capacity to offer quality training programs; (ii) a TA for IT workforce gap analysis, development of digital job taxonomy/skills framework, and improving digital talent recruitment, retention, and reskilling within the civil service; (iii) training of trainers (teaching staff of relevant HEIs) programs on curriculum development, pedagogical design, teaching methods, and innovative pedagogical technologies; (iv) delivery of training programs for targeted public agent users to improve the uptake of existing digital public platforms such as IFMIS, HMIS, EMIS, SYDONIA World, distance learning platform, etc.; (v) delivery of training programs for specialized digital skills in selected MDAs; (vi) hands-on support for the simplification and business process reengineering for the selected digital public services under Sub-component 3.4.; and (vii) strengthening digital skills of government staff to increase adoption of digital government services, that includes training content on using digital tools to deliver online government services to citizens and support business continuity during times of adverse weather conditions and extreme climate events⁷⁶.

Subcomponent 2.3 Digitization of payments (US\$10 million equivalent)

38. Proposed interventions under this subcomponent will support the development of DFS through greater accessibility to and usage of financial products and services for government employees as well as the whole population, including women and marginalized populations. In fragile countries like Chad where instability and limited resources are jeopardizing the achievement of global development targets, inclusive finance plays a role in bolstering resilience and unlocking opportunities for low-income people⁷⁷. This sub-component aims to increase financial inclusion, improve transparency and accountability, and enhance the government's monitoring capabilities. Chad being a member of the

⁷⁴ Procurement/Acquisition of all ICT equipment for the CERT, SOC, data protection and digital certification system or PKI will be certified by Energy Star or equivalent standards, which would surpass the energy efficiency levels of equipment existing in the country. There are no mandatory energy efficiency standards or labeling schemes for energy efficiency in Chad (Source: [ADERM](#)).

⁷⁵ Online violence can reduce women's voice or have psychological impacts, with women reporting anxiety, panic attacks, and concerns about online violence escalating into physical or sexual violence. Perceived risks around online activity may limit women's willingness to explore digital jobs or participate in the digital economy.

⁷⁶ Delivery of government service online is expected to improve business continuity and uninterrupted services delivery for government, citizens and firms, when a natural disaster strikes.

⁷⁷ Financial inclusion can be a significant factor in addressing conflict and violence in fragile areas. By providing individuals and communities with access to financial services, such as savings, credit, and insurance, financial inclusion can help to reduce poverty and create economic opportunities. This can lead to improved livelihoods and can help to address some of the underlying causes of conflict and violence, such as economic disparity and lack of access to resources. Additionally, financial inclusion can help to integrate marginalized groups into the formal economy, which can promote social cohesion and stability.



CEMAC, the proposed activities will require coordination with other regional initiatives regarding aspects relating to the regulatory framework and payment systems. The proposed activities include:

- (a) **The acceleration of the digitalization of government payments with the development of technical interfaces and synergies between the different institutions involved, including the treasury and the central bank (*Banque des États de l'Afrique centrale, BEAC*) (US\$2 million equivalent).** It notably aims to (i) promote P2G and Business-to-Government electronic payment streams by implementing a single gateway to foster government acceptance; (ii) establish synergies between government entities and institutions on revenue collection payment processes, thus streamlining the sourcing and operation of such services; and (iii) enhance the digitalization of G2P and G2B transactions, including the disbursement of salaries, pensions, social benefits, subsidies, and contract payment. TA will be included to: (i) enhance capacity of the treasury and other public institutions to monitor and process electronic transactions; (ii) support public institutions in their electronic payment roadmap and in their communication with citizens regarding the delivery of government payments for selected use cases, such as the digitalization of pensions and social safety nets⁷⁸ (including cash transfers⁷⁹). Specific use cases might require enhanced arrangements, embedding flexibility, to ensure fragile individuals are sustained appropriately and are guided toward more convenient interaction with the government.
- (b) **The incentivization of DFS to strengthen financial inclusion (US\$3 million equivalent),** with technical assistance to: (i) better map and analyze the distribution of financial access points leveraging geo-spatial technology in order to design policies aiming to bring financial access points — such as bank branches, mobile-money agents, and ATMs — close to the population, especially in rural and conflict-affected areas, and climate hot-spots⁸⁰; (ii) update key policy documents relating to consumer protection and financial inclusion, including the update of the 2016 national financial inclusion strategy; (iii) establish a consumer protection framework (observatory or ombudsman) as aligned with the regional CEMAC Consumer protection Law and strengthen capacities; (iv) develop a National digital and financial literacy Strategy (including gender-specific barriers) and implement related programs; and (v) support to the development of a conducive merchant payments ecosystem. Incentivizing digital financial services will provide a cost-efficient way to reach the most vulnerable (users of DFS and particularly those in the social safety net programs) with financial services which will timely and quickly help strengthen their resilience and better prepare for any hazard event, especially a climate one (for instance, in the aftermath of floods/drought through quick electronic cash transfers via mobile money). The project will consider complementary interventions designed to close DFS and financial inclusion gender and exclusion gaps, including digital platforms specifically designed for low-literacy populations, such as image-based mobile-money agent applications that increase the transaction transparency for low-literacy clients as well as coupling digital skills training with financial literacy training for women.⁸¹
- (c) **The strengthening of the framework for Anti-Money Laundering and Combating the Financing of Terrorism (AML/CFT) (US\$5 million equivalent)** by (i) increasing the country's understanding of AML/CFT risks through the dissemination of the findings of the NRA report; (ii) supporting the preparation of a national AML/CFT strategy; (iii) strengthening existing economic and financial intelligence networks; (iv) providing technical assistance to enhance Chad's financial sector supervisors' capacity to improve the prevention and the detection of illicit financial transactions

⁷⁸ Digitally depositing social safety net payments into women's accounts increases privacy and control over use:

<https://www.journals.uchicago.edu/doi/full/10.1086/687578>

⁷⁹ Under the Chad Adaptive and Productive Safety Nets Project (P502142) some cash transfers programs will be digitalized. This activity will complement the digitalization of such programs. The electronic shock-responsive emergency cash transfers will allow quick and timely response to the beneficiaries of Chad Adaptive and Productive Safety Nets Project that provides cash transfers supporting rapid flood response mechanism.

⁸⁰ Climate hot-spots here refer to those areas that are increasingly/frequently impacted by floods and droughts and need response and recovery support from the government

⁸¹ Digital platforms and interface design should be inclusive and co-developed with the relevant marginalized populations;

<https://www.anc.edu.ro/wp-content/uploads/2021/04/Designing-Inclusive-Digital-Solutions-and-Developing-Digital-Skills.pdf>



in the private sector and the enforcement of administrative sanctions for non-compliance with AML/CFT obligations; and (v) TA to strengthen actions for financial inclusion and transparency to prevent the use of informal financial services, including for money transfers.

Sub-component 2.4 Digitalization of selected services (US\$9 million equivalent)

39. **This sub-component aims to facilitate the digitalization of selected, high-impact, and demand-driven services.** Sectors identified include education, and public procurement, which are considered priorities by the GoC. Modernization of systems would be done based on the IT master plans developed under Subcomponent 1.2 and taking into account the fact that the country does not have an energy efficiency standard. Financing would include software, related hardware⁸², and TA necessary for all selected services.

- (a) **The digitization of the administrative procedures and processes for education degree certification (US\$2 million)**, through: (i) a TA for the optimization of administrative procedures and processes for education degree certification; (ii) digitization of education degree certification records (archives); (iii) small rehabilitation works for the National Directorate of Exams and Competitions (*Direction Nationale des Examens et Concours*); and (iv) acquisition of hardware and software, including customized information systems for the education degree certification and an Enterprise Resource Planning and Student Life Cycle System (for the three education ministries: Ministry of National Education and Civic Promotion (*Ministère de l'Éducation Nationale et de la Promotion Civique*); Ministry of Higher Education, Scientific Research, and Innovation (*Ministère de l'Enseignement Supérieur, de la Recherche Scientifique et de l'Innovation*); and Ministry of Vocational Training and Jobs (*Ministère de la Formation Professionnelle et des Métiers*)).
- (b) **The deployment of an electronic Government Procurement (e-GP) solution (US\$7 million)** through financing for : (i) the development and deployment of the selected e-GP system solution; (ii) TA (just-in time) to support e-GP expansion and the national capacity-building plans; (iii) change management activities, training and communication activities for e-GP implementation; and (iv) IT equipment⁸³ for the deployment of the e-GP project.

Component 3: Project Management and knowledge transfer (US\$6 million equivalent)

40. **This component would finance project management costs and institutional coordination, including setting up a dedicated Project Implementation Unit (PIU) and coordination mechanisms.** It includes procurement, financial management (FM), monitoring and evaluation (M&E), environmental and social (E&S) management and citizen engagement support, inclusive consultation mechanisms, and qualitative user research and satisfaction surveys as necessary, as well as communications. It would also seek to strengthen the capacity of implementing MDAs, ensure quality assurance to ensure adherence to best practices on procurement related to technology, facilitate on-the-job learning, transfer competencies and advisory services needed. The transfer of knowledge between the PIU experts and national institutions will be achieved through shadow management, ad hoc south-south exchanges, and the use of country systems whenever possible (e.g., Universal Service Funds under Component 1).

41. **Cross-cutting Citizen Engagement (CE) support will be financed under this component.** This includes inter alia the development of a comprehensive CE strategy and fit-for-purpose consultation mechanism to ensure that all relevant stakeholders, including marginalized groups and stakeholders at risk of exclusion, are consulted about the project

⁸² All hardware and ICT equipment under this subcomponent will be certified by Energy Star or equivalent standards, which would surpass the energy efficiency levels of equipment existing in the country.

⁸³ All hardware and ICT equipment under this subcomponent will be certified by Energy Star or equivalent standards, which would surpass the energy efficiency levels of equipment existing in the country.



implementation progress on a rolling basis (e.g., satisfaction survey), and in real time through the GRM system in place (see paragraph 42). CE interventions are streamlined across components as the Project will systematically include consultations and user research, as well as feedback gathering on all project-financed infrastructure, training programs, systems, and services. The feedback from those consultations is fed back into the implementation of project-financed activities. Project beneficiary feedback will be translated into corrective actions reflected in the AWPB, approved by the Project Steering Committee.

42. **The establishment of a comprehensive and accessible Grievance Redress Mechanism (GRM) with adequate procedures and staffing will be supported under this component to allow for receiving and responding to stakeholders' and beneficiaries' grievances, as well as any feedback on project-financed activities, within set standards and timeframes.** The GRM will leverage existing national and local GRMs, including links with the project M&E system to enhance sustainability, avoid duplicative processes, facilitate a smooth integration of project activities, and improve project monitoring data. More specifically, the Project will finance goods and consulting services to (i) develop and implement tools to support better collection, tracking, and response to grievances; (ii) provide capacity building for PIU and MCENDA personnel; and (iii) develop and implement feedback mechanisms to ensure that GRM data are used continuously to improve project implementation and results. The results framework has a CE indicator which will track those grievances, feedback, and ensure that they are addressed/taken into account in a timely fashion.

Component 4: Contingent Emergency Response Component (CERC) (US\$0 million)

43. **A CERC is included in the project per the Investment Project Financing Policy, paragraphs 12 and 13, for Situations of Urgent Need of Assistance and Capacity Constraints.** This will allow for rapid reallocation of project uncommitted funds in the event of an eligible emergency as defined in OP 8.00. A CERC Manual will guide the activation and implementation of the CERC, and an Emergency Action Plan will be prepared to confirm activities and financing for a specific event.

C. Project Beneficiaries

44. **Project interventions related to expanding digital connectivity and facilitation of the digital delivery of public services will positively impact a wide range of beneficiaries, especially in remote and fragile areas.** Individuals, in particular women, youth, PwDs, refugees, IDPs, businesses, and government agencies will receive improved access to broadband and digital services. The project adopts a spatial approach, considering climate risk, security risk, energy access, ongoing World Bank-financed initiatives (for example, in the energy, transport, social protection, health, education sectors), and population density. The country economy will benefit not only from an increased mobilization of private investment in the digital sector, but also from digital skills literacy programs that will facilitate the productive use of internet and boost employability for the youth.

- (a) **Individuals.** The project will lead to the delivery of better and more affordable broadband internet to more than 4.5 million people, especially those in unserved or underserved areas, marginalized or vulnerable communities: women and girls, PwDs, IDPs and refugees, and communities in areas of high climate risk. In addition, the project enable access to digital services to 2 million additional people across the country. Some 40,000 beneficiaries will directly receive digital trainings —comprising at least 25,000 women and girls and 2,000 PwDs.
- (b) **Government institutions.** The direct beneficiaries of the project include MCENDA, MFBEP, Ministry of Training and Financial inclusion, ARCEP, ADETIC, ANSICE, and ANATS. More generally, public institutions across government will benefit from improved internet connectivity, shared digital infrastructure, new and improved G2G services, digital payments, IT equipment, and digital skills trainings – including the Ministries of Education, Higher Learning, and



Health. MDAs across government will benefit from an improved ability to manage data in a more secure and cost-effective manner. At least 5,000 civil servants will be trained in management or use of digital systems and infrastructure. Many decision-makers in MDAs will benefit from training and shadowing on relevant expertise.

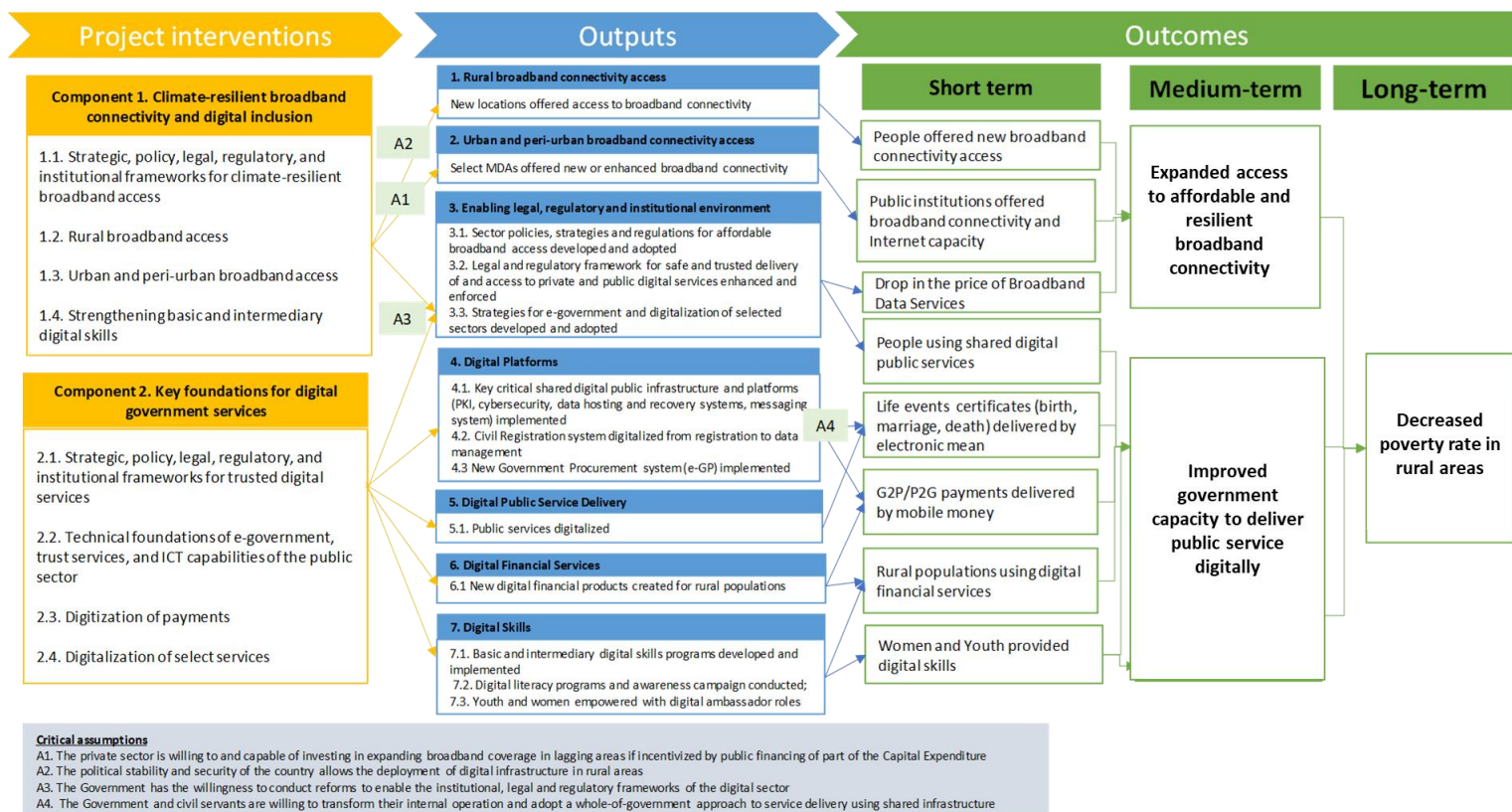
- (c) **Private sector.** The telecommunications sector, including mobile network operators and internet service providers, will broadly benefit because the project will enable a reduction in the costs of network deployment in challenging areas and promote a more competitive market, and a secure environment for online operations. FSPs, including banks and other payment service providers such as mobile money operators, as well as merchants, will also directly benefit from the efficiency gains generated by the transition from cash-based transactions to DFS. Other businesses are indirect beneficiaries, who will benefit from the reforms supported, including a secure environment for online services, wider network coverage, and potential downstream G2B e-services enabled through project investments.

D. Results Chain

45. **The project's theory of change reflects the key assumptions, challenges, and solutions captured in the proposed project design.** Component 1 seeks to address the stark digital divide by stimulating private sector investment in digital infrastructure and last mile access to digital services, as well as improve connectivity of MDAs nationally, beyond N'Djamena. Component 2 finances hardware, software, and TA to help lay digital government foundations and establish secure digital platforms and services. Component 3 supports project management and ensures that capacity is developed adequately throughout public institutions involved in the project to ensure sustainability of project outputs. This is based on key assumptions that no dramatic shocks will change the economic and fiscal environment, that the Government will remain committed to supporting competition and adopt the proposed legal and regulatory frameworks, that the private sector will be willing to expand provisions of digital services, and that beneficiary populations will actively participate in the proposed digital access initiatives when sensitized to the benefits of digital.



Figure 3 : Chad Digital Results Chain



E. Rationale for World Bank Involvement and Role of Partners

46. **Public sector financing is appropriate as it will be used to catalyze private sector investment and support public goods.** The World Bank is well-positioned to support this sector, given past analytical work and its track-record of implementing similar operations in the region and beyond. The World Bank is thus able to leverage global expertise in areas spanning broadband market development and e-government. The project design has been informed by global and country analytical work. This has provided the World Bank with a solid understanding of the challenges and the development of an effective working relationship with the stakeholders, positioning the World Bank as a strong ecosystem convener.

47. **World Bank financing is appropriate as it will be used to (i) create an enabling environment conducive to unlocking private sector investment; (ii) address market failures; and (iii) introduce digital public goods.** The project will stimulate private sector development through TA for upstream legal and regulatory reform that can enable scaled investment in broadband market development and e-services. Public financing will only be employed to the extent necessary to reach areas where private sector investment incentives are not adequate to increase access to broadband. The project will also focus on digital public infrastructure for building government's ability to deliver services digitally; tapping into the commercially-off-the shelf solutions which are pre-built software or hardware components that can be easily integrated into existing government systems. By using these solutions, the project could save time and resources and focus on customizing the components to meet the specific needs of the government.



48. **The World Bank is well positioned to support Chad's digital development, given past investments in the ICT sector as well as recent analytical works completed.** Activities financed are informed by the recently completed DE4A country assessment that identified several binding constraints but also opportunities associated with accelerated digital adoption. This has allowed the World Bank to emerge as a trusted partner and leading development actor and convenor in this space. As detailed in section F, the World Bank is also able to draw on significant regional experience in implementing similar operations regionally and beyond.

49. **The project complements support provided by other financial partners, thereby capitalizing on existing interventions, spatial approaches and lessons learned.** The World Bank is set to build synergies with several development partners; examples include (i) collaboration with AfDB on the strengthening of the digital economy strategic framework and broadband connectivity expansion; (ii) collaboration with the EU on digitizing public procurement and with the European Investment Bank for potential parallel financing synergies on the digitalization of the public administration and services; (iii) collaboration with the EU and UNICEF on digitizing and modernizing CR as well as scaling-up the coverage; and (iv) collaboration with UN agencies and GIZ on digital inclusion initiatives.

F. Lessons Learned and Reflected in the Project Design

50. **For Chad -- an FCV country, volatile security situation -- lessons learned from Burkina Faso, Mali and Niger point to the need to adopt flexible implementation strategies.** An "on-the-ground" presence of Bank staff coupled with a close monitoring of the security situation is necessary. Appropriate and timely actions need to be taken in coordination with the security team in the Country Management Unit. Also, security risks should be considered upfront in the design of the project to ensure the availability of digital services in case of a security incident (e.g., fiber optic link redundancy, satellite backup for connectivity, data hosting redundancy, etc.).

51. **The project takes a comprehensive approach to stimulating resilient broadband market development and digital adoption.** Connectivity elements of the proposed project leverage lessons learned from a comparative analysis of 70+ last mile connectivity deployments, which points to the need for designing infrastructure tenders in projects using technology neutral principles to maximize competition, value for money, and scope for using innovative business models for network expansion. The project also takes a comprehensive approach to stimulating demand, by addressing all key access barriers linked to affordability, readiness, and relevance, where more novel areas such as device affordability draw on research conducted by and with partners such as GSMA⁸⁴ and global experience of smartphone subsidy schemes⁸⁵, including related interventions in social protection projects.

52. **The project pairs interventions that support a whole-of-government approach to government digitization with investments that mitigate growing risks associated with increasing digital adoption.** This project benefits from lessons learned from many countries⁸⁶ on broad digital governance policy and institutional reforms. Interoperability is one of the key enablers for a whole-of-government approach. Specific World Bank-financed projects⁸⁷ that demonstrated successful implementation of e-GP (e-Government Procurement) systems and reforms have also provided valuable lessons. An

⁸⁴ GSMA. 2017. "[Accelerating Affordable Smartphone Ownership in Emerging Markets.](#)"

⁸⁵ Examples include: full price of a device called Ascend Y21 was reverse subsidized by MTN Ghana in the form of voice and data services over a 12-month period; Columbia allocated US\$90 million for data and smartphones subsidies for low-income households; Malaysia launched a national program to encourage youth to purchase 3G-enabled smartphones with a rebate on certain phones reducing the cost by 40 percent.

⁸⁶ These include the Uganda Digital Acceleration Project - GovNet (P171305); Albania Citizen-Centric Service Delivery Project (P151972), eTransform Ghana Project (P144140), Modernization of Government Services in Republic of Moldova Project (P148537), Rwanda Digital Acceleration Project (P173373); Mozambique Digital Governance and Economy Project (P172350); and Caribbean Digital Transformation Project (P171528).

⁸⁷ Including the Rwanda Public Sector Governance Program-For-Results (P149095); Digitizing Implementation Monitoring and Public Procurement Project in Bangladesh (P160758); Maghreb Public Procurement Reform Advisory Services and Analytics (P171637) and Morocco Public Sector Performance Program-For-Results (P169330).



impact evaluation⁸⁸ of e-procurement adoption in Bangladesh demonstrated that e-procurement reduced integrity risks, increases participation of firms and reduces prices offered by firms during the bidding process. While technical issues are important, the success of e-GP or any digital systems largely depends on effective management, training, and stakeholder engagement. All the learnings are reflected in the sequencing of subcomponents under Component 2 activities, especially those related to legal, regulatory, and technology frameworks, the development of digital public infrastructure, platforms and services, digital interoperability, and digital change management and usage.

53. **Digital skills training approach draws on existing successful initiatives**, such as the digital ambassadors in Rwanda and other impactful experiences (for example, from Nigeria, Burkina Faso, and Sierra Leone). Experience with digital centers such as in the Niger: Smart villages for rural growth and digital inclusion project (P167543) have also been drawn upon. Lessons learned from EdTech initiatives have also showed the importance of careful planning, stakeholder engagement, infrastructure, teacher training, curriculum alignment, monitoring, and sustainability in the successful implementation of EdTech initiatives.

III. IMPLEMENTATION ARRANGEMENTS

A. Institutional and Implementation Arrangements

54. **Strategic project oversight and governance will be provided through a Project Steering Committee (PSC) that has been established through a Prime Minister decree *Arrêté 0748/PT/PM/2024* dated March 21, 2024.** The PSC oversees the implementation of this multi-sectoral project and advises the PIU. Its mandate includes, inter alia: (i) reviewing project progress on an annual basis (additional meetings can be scheduled as needed); and (ii) provision of strategic guidance and recommendations to the PIU and Technical Committee related to project implementation and/or any project restructuring needed. The PSC is chaired by the MCENDA with the MFBEP acting as Vice Chair, and the PIU Coordinator acting as its secretary. The PSC comprises representatives from all implementing ministries. The Terms of Reference (ToR) for the PSC will be further detailed in the Project Implementation Manual (PIM), to be prepared by the Recipient before effectiveness.

55. **A PIU will be established within MCENDA, which will serve as the main implementing agency.** MCENDA is well-placed to serve as the institutional anchor for the PIU, given its mandate to develop Chad's digital economy. The PIU will lead project implementation and facilitate daily collaboration with project stakeholders. It will also oversee core project-related fiduciary functions, including FM and procurement, as well as project-related M&E, E&S commitments, and will consolidate and submit project reporting to the PSC, as well as to the World Bank. Key staff, including the Coordinator, Deputy Coordinators/Technical Leads (including infrastructure, payments, e-government experts), Senior FM Specialist, Senior Procurement Specialists, a social specialist, and an environment specialist will be hired using a Project Preparation Advance prior to effectiveness. A digital infrastructure technical lead; a digital payments technical lead; an e-government technical lead; and (d) an internal auditor will be hired no later than three months after effectiveness. The technical leads will act as deputy coordinators in charge of the implementation of activities under their purview. The detailed composition and role of the PIU will be set forth in the PIM.

B. Results Monitoring and Evaluation Arrangements

56. **A Results Framework has been developed to track progress towards achieving the PDO, and the PIU will be responsible for reporting on indicators.** The PIU, in consultation with partner MDAs, will prepare progress reports every

⁸⁸ World Bank, e-Procurement Reform in Bangladesh – case study, available here: <https://pubdocs.worldbank.org/en/115291611673413993/e-Procurement-Reform-in-Bangladesh.pdf>; and see also: <https://www.worldbank.org/en/results/2023/05/03/world-bank-helps-bangladesh-transform-public-procurement-into-a-sustainable-online-system>



semester, for submission to the World Bank no later than 45 days after the end of each calendar semester, in accordance with the agreed format. Related reports will cover (i) physical and financial progress achieved against agreed implementation and disbursement indicators; (ii) status of procurement of activities; (iii) issues and problem areas, including comments on actions to address identified problems; (iv) work programs and budget, including forward-looking estimates; and (v) progress in PDO and intermediary indicators.

57. **For real-time data collection and analysis, the Project will rely on the Geo-Enabling Initiative for Monitoring and Supervision (GEMS) or hire Third-Party Monitoring (TPM) where relevant.** GEMS enables project teams to use open-source tools for in-field collection of structured digital data that automatically feeds into a centralized M&E system. Data sources can include tailor-made forms, photos, audio, videos, time and date stamps, and Global Positioning System (GPS) coordinates that allow for automated geo-mapping of the information. Using these tools systematically makes it possible to enhance the transparency and accuracy of M&E and increase the accountability of monitoring. Also, it provides a platform for remote supervision, real-time safeguards monitoring, and portfolio mapping for coordination across projects and partners. Relevant PIU and technical stakeholder representatives will be trained to benefit from the extensive experience gathered through GEMS in other projects. Given the nature of the investments and the changing dynamics on the security aspects, the use of TPM, especially in areas that are difficult to reach or where the security situation is deteriorating, will be considered.

58. **In-built systems for tracking results and satisfaction surveys will be leveraged to support stakeholder engagement and solicit beneficiary feedback periodically.** Related tools will be embedded directly into project delivery to ensure feedback in real time, using digital tools and systems to register beneficiaries and report their feedback (e.g., using tablet-based or rapid mobile/short message service survey tools). Beneficiary focus groups will also be leveraged to inform design and track progress over time.

59. **Working with the World Bank's Africa Gender Innovation Lab, the project will also explore including a rigorous impact evaluation.** A potential impact evaluation would prioritize innovative activities designed to boost women's access and control of digital devices, digital skills, or use of digital services, and will depend on the key binding constraints facing women. For example, if restrictive intra-household or community norms are particularly relevant in holding back women's access and control of digital devices, the project may consider a norms-based intervention, couples' trainings, or soft-skills negotiation-focused training. Alternatively, if liquidity or credit constraints are particularly binding and preventing women from being able to afford devices, the impact evaluation could focus on digital device loans. Similarly, the project may evaluate digital services which have the potential to ease a wide range of constraints: well-designed digital financial services—for example leveraging digital payment transactions to facilitate cash flow-based lending—could ease credit or collateral constraints; ag-tech services for farmers could address access-to-market constraints; and WhatsApp groups of entrepreneurs could ease information or network constraints. An impact evaluation under the project is subject to funding and the identification of policy relevant research questions as well as the feasibility of a suitable rigorous research design.

C. Sustainability

60. **Sustainability will primarily be ensured through (i) prioritization of energy-efficient and climate-resilient approaches and solutions; (ii) use of private sector-led and market-based mechanisms for resource allocation and investments in telecommunications infrastructure to expand broadband access; and (iii) investment in digital public infrastructure that can reduce the long-run costs of public service delivery.** While many international firms/contractors will be hired to support the various technical activities envisioned, heavy emphasis will be placed on knowledge transfer to ensure that local teams are able to maintain systems and solutions introduced.



IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis

61. **Benefits of Chad Digital Transformation, that are expected to be substantial, come through three axes: (i) expanded broadband access and provision of basic and intermediate digital skills; (ii) advanced digital skills training to civil servants and delivery of digital public services; and (iii) increased financial inclusion.** On broadband access, the project is expected to crowd in private investment at least equal to the public funding and offer broadband access to far-to-reach populations through the deployment of 500 fully operational sites, each site covering around 3,000 inhabitants on average and costing US\$160,000 for installation and commissioning (CAPEX). Also, basic, and intermediate digital skills training will be delivered to at least 40,000 potential users of mobile broadband services. Concerning the digitalization of public services, the project intends to lay the foundations of digital government, modernize the civil registry system, and support the certification of at least 6,000 civil servants in advanced digital skills. Concerning the financial inclusion, the project aims to promote the use of mobile money for G2P/P2G payments and digitalize selected sectors. Given the available data, several assumptions have been made to make the calculation of net present values and internal rates of return possible. However, a lot of benefits are not quantifiable.

62. **The project is designed to catalyze commercial financing in alignment with MFD principles.** Consultation with private operators suggests that they are ready to participate in any commercial opportunities created by the project. The project will seek to promote private sector investment through (i) upstream policy, legal, and regulatory reforms that create a more enabling environment for broadband, DFS, ICT services, and equipment providers to invest, by revising existing frameworks as well as bridging gaps identified; (ii) digital access initiatives that help stimulate demand and create a growing user base that can increase and sustain commercial investments incentives (for example, by supporting digital literacy and subsidizing devices); and (iii) investment in shared infrastructure and platforms that serves as digital public infrastructure and can help reduce transactions costs and increase market efficiency and innovation (for example, via investment in core payments infrastructure, authentication, and cybersecurity). The World Bank will be collaborating closely with the International Finance Corporation (IFC) to unlock prospects for downstream investments in the digital infrastructure.

Broadband access and use

63. **The rural connectivity sub-component is viable financially and economically beneficial.** The results for the project without public capital support are: (i) the financial rate of return (FRR) (C – without capital) is 34.59 percent; (ii) the break-even point occurs in 12 years; and (iii) the Net Present Value (NPV) is US\$ -194.67 million (with a discount rate of 10 percent). These results show that although FRR (C) is positive, telecommunication operators would not invest in this project alone because the break-even is far away. Therefore, public financing is needed to make the project possible.

64. The results for the project with public capital support are: (i) the FRR (K – with capital) is 45.86 percent; (ii) the break-even point occurs in eight years; and (iii) the NPV is US\$114.13 million (with a discount rate of 10 percent). These results show that the project becomes acceptable for private operators. It refers to the point at which total costs and total revenue generated by the deployment of their networks are equal, meaning that a business is neither making a profit nor a loss.



Table 3. Economic analysis of the benefits of investments in rural connectivity

Broadband Access	Public CAPEX (US\$ Million)	Private CAPEX (US\$ Million)	Private OPEX (over 20 years) (US\$ Million)	Total Cost (over 20 years) (US\$ Million)	Total Revenue (US\$ Million)	Financial Rate of Return (20 years) (%)	Break-even point (Year)	NPV (with 10% Discount rate) (US\$ Million)
Without Public Capital		67.00	275.04	355.04	477.86	34.59%	12	194.67
With Public Capital	40.00	40.00	275.04	355.04	477.86	45.86%	8	114.13

65. **On the economic side, benefits of World Bank investments to expand broadband access in rural areas (US\$40 million) and in urban and peri-urban areas (US\$11 million) for the connectivity of the MDAs, are expected to be substantial.** According to a study carried out by the ITU⁸⁹, an increase of 10 per cent in mobile broadband penetration yielded an average increase in 1.5 per cent in GDP in African countries. Improvements in the regulatory environment and provision of targeted subsidies to commercial operators/Internet service providers to expand broadband infrastructure in Chad under the project is expected to increase “Unique” Mobile Broadband users in the country by 8.58 percentage points during the project period. This is equivalent, for the whole national economy, to an increase of the GDP of US\$170.77 million, more than the US\$67 million to be invested in rural area (US\$40 million of public capital and US\$27 million expected from private actors) and the US\$14 million to be invested in urban area (US\$11 million of public capital and US\$3 million expected from private actors).

66. **Public financing for broadband connectivity expansion and upgrades (under Subcomponent 1.2 & 1.3) is expected to mobilize approximately US\$30 million (US\$27 million in rural areas and US\$3 million in urban areas) in direct private sector investment in broadband infrastructure.** This assumption is made from lessons learned in Niger (Smart Villages Project for Rural Growth and Financial Inclusion – P167543) which is a country like Chad and Ghana (E-Transform Ghana Project - P144140). While comparative schemes in Tanzania and elsewhere have mobilized more the extremely low ARPU (around US\$4 per month) and the limited access to supporting infrastructure, such as energy, explain the more conservative estimates.

67. **Benefits of the investment in digital skills cannot be fully measured.** However, we have valued an additional turnover for mobile operators related to the increase in broadband users and the productivity gain of civil servant consequences of their advanced digital skills. If we consider that half of the individuals provided basic or intermediate digital skills will become active broadband user and a productivity gain of 20 percent for civil servant receiving advanced digital skills trainings, the project can generate, during its lifetime, enough benefits (US\$10 million) to cover the US\$5 million investments in digital skills. The NPV would be US\$1.64 million with an internal rate of return (IRR) at 52 percent for a discount rate of 10 percent.

Digital public service delivery

68. **Accelerating the digital transformation of public services is expected to generate significant additional benefits that cannot be quantified due to lack of data.** The most significant are described below:

- (a) **Reduced time and travel costs for people and businesses accessing services.** Expanding broadband to MDAs will reduce the time and travel costs of people newly gaining access to digitally enabled public services. In addition, people who would otherwise forgo the opportunity to gather information or access a service because of prohibitively high time and travel costs will now be able to access services, leading to their improved health and well-being and higher productivity, among other things.

⁸⁹ https://www.itu.int/hub/publication/d-pref-ef-bdt_afr-2019/



- (b) **Increased access to public services for people and businesses.** Reengineering and simplifying the procedures and processes of digitalized public services and integrating them into the shared platforms will help boost both demand for and access to public services, strengthening the social contract. Expanded access to public services is also expected to be especially beneficial to women, marginalized people, and PwDs, as they are less likely than men to access services due to the high time and travel costs involved in paper-based systems.
- (c) **Through the adoption of an e-GP system, for instance, the project will generate significant cost savings for the public administration.** In other countries, savings can represent 5–8 percent of the administration's annual procurement budget, as a result of increased competition, transparency, governance, and efficiency. By automating all aspects of government procurement from planning, advertising, and tendering to contract signing and, the first phase of e-GP will ensure continuity in government operations and delivery of services to people, particularly in the context of health pandemics or national emergencies. Benefits to the private sector due to improved transparency of information relating to tenders include an increase in the number of private sector entities, particularly local Small and Medium-sized Enterprises (SMEs), who bid for government procurement opportunities and have access to public contracts, thus improving competition, and contributing to job creation. Improved transparency and integrity will also contribute to reducing corruption and enhancing people's and the private sector's confidence in the government procurement process and the utilization of public funds. A more accurate predictive calculation of economic benefits will depend on the adoption/transition model chosen (Custom vs. Commercial off-the-Shelf vs. Software-as-a-Service), in particular as to the expected benefits in the first years of the project, until the e-GP system covers all public buyers.

Financial inclusion

69. **The proposed activities relating to the digitalization of government payments and the promotion of DFS are expected to generate substantial benefits in terms of financial inclusion**, while promoting innovation and competition in the field of payments and financial services more generally.

70. **The digitalization of government payments can notably yield a multitude of advantages for both public authorities and recipients.** From the government's perspective, it allows to generate efficiency gains and to reach a broad population in a rapid and secure way, while curtailing the potential for funds to be siphoned off or misused during cash distribution, thereby reducing leakages and corruption. For recipients, government payments serve as a gateway to financial inclusion and economic empowerment, particularly benefiting marginalized groups such as women and young people. As they represent large volumes, the digitalization of important government payment streams – such as public pensions, public salaries, or social benefits – can be a catalyst for the end-to-end automation of public processes related to payments and, more broadly, for the development of DFS in the country.

71. **Through other interventions, the project will also reinforce the enabling environment for the development of electronic payments and DFS.** By strengthening consumer protection and financial literacy, developing access points across the country, and supporting the roll-out of innovative use cases and practices, it aims to incentivize usage of electronic means of payment as a first step towards financial inclusion. In the longer term, it should bring wide-ranging benefits to individuals and businesses, promoting the development of a digital economy, facilitating day-to-day living for families, and helping people to plan long-term goals or cope with unexpected emergencies.

Paris Alignment

72. **The operation is aligned with the goals of the Paris Agreement on both mitigation and adaptation**

73. **(a) Assessment and reduction of adaptation risks.** High temperatures and floods, as identified by the Climate and Disaster Risk Assessment, are the main climate risks likely to affect the project investments, namely the digital



connectivity deployment (e.g., damage to cables and service disruption due to floods), refurbishment/construction of digital user center and improvements in government's data. To manage these risks, the project will conduct site-specific climate risk assessments, and integrate adequate adaptation measures to strengthen climate resilience of digital connectivity financed under the project (see details of climate adaptation measures under Components 1 and 2). To ensure service continuity and energy supply following flooding events, batteries charged from solar panels will be considered. Data hosting arrangements will also incorporate climate risk considerations, including disaster recovery strategy to avoid loss of data due to service interruption caused by extreme weather events such as intense rainfall and floods (see details on climate resilience measures under Subcomponent 2.2). The refurbishment/construction of digital user centers will include site-specific climate risk assessment to determine the need for adaptation measures, and adequate measures will be integrated accordingly. Altogether, these measures will ensure that the adaptation risks are reduced to an "acceptable" level.

74. **The project will contribute to the country's low-carbon development goals** by prioritizing renewable energy to power the digital connectivity infrastructure (e.g., by powering cell-sites with solar) and by investing in energy efficient telecommunications infrastructure and equipment. The TA on data center and improvements in government's data hosting arrangements will follow internationally recognized best practices on energy efficiency⁹⁰ to reduce the mitigation risk to a 'low' level. A site-specific preliminary assessment will be conducted to identify energy efficiency measures for the refurbishment/construction of digital user centers and best-possible lower GHG/energy efficiency measures (that are economically and technically feasible) will be incorporated. Where there is no grid connection available for digital user centers in rural areas, renewable energy would be utilized. The IT equipment (computers, printers, scanners) and other electrical equipment such as air conditioning units will be certified energy efficient. These measures together will ensure that the operation will not increase the danger of carbon lock-in in the long term and it has a low risk of preventing the country's transition to low-carbon development pathways.

B. Fiduciary

75. **An FM assessment was conducted to assess the adequacy of the FM arrangements for the Chad Digital Transformation Project.** Project overall FM activities will be executed by a new PIU to be established under the MCENDA. The assessment was conducted in accordance with the FM Manual for World Bank Investment Project Financing Operations issued on September 7, 2021. **The assessment concluded that the FM overall residual risk is Substantial**, mainly because the implementation of the project will include various technical groups from ministries other than the PIU. In addition to this, the assessment revealed that the project implementation entity does not have the necessary material and human resources to ensure financial management of the project in accordance with World Bank requirements (lack of relevant experiences in the management of projects and programs financed by the World Bank, lack of qualified financial management staff, lack of accounting software, and manuals of accounting procedures and financial management; and weak internal control).

76. **To mitigate the FM risks**, the PIU of the ongoing Sahel Women's Empowerment and Demographics Project (SWEDD-TD⁹¹ P150080) will ensure the financial management aspects of the project until the full establishment of the new PIU. The current implementation arrangements in place for the SWEDD-TD financial management will be applied for this project. To build the FM capacity and system of the MCENDA-PIU, several measures will be considered by project effectiveness: (i) the recruitment and onboarding of an experienced FM officer and an Accountant; and (ii) the elaboration and adoption of the PIM, including the FM procedures manual. Additional measures include: (i) the acquisition and installation of an appropriate accounting software, parameterized to handle accounting and reporting needs, three months after effectiveness; (ii) the recruitment of an internal auditor to conduct ex post reviews of the project transactions

⁹⁰ Such as 2019 Best Practice Guidelines for the EU Code of Conduct on Data Centre Energy Efficiency (JRC).

⁹¹ "TD" is the ISO country code for Chad.



and procedures, and to ensure that identified weaknesses are addressed in a satisfactorily manner within three months after effectiveness; and (iii) the recruitment of an external auditor to conduct an annual financial audit of the financial statements of the project, and a review of the internal control system, six months after effectiveness. (See Annex 1 for more details).

77. **Procurement under the proposed project will be carried out in accordance with the World Bank “Procurement Regulations for Investment Project Financing Borrowers” dated September 2023**, the ‘Guidelines on Preventing and Combating Fraud and Corruption in Projects Financed by IBRD Loans and IDA Credits and Grants’ dated October 15, 2006 and revised in January 2011 and as of July 1, 2016, and beneficiary disclosure requirements, as well as other provisions stipulated in the project Legal Agreements. The overall procurement risk of the project is significant. The procurement capacity of all entities involved will be strengthened during implementation.

78. **Project Procurement Strategy for Development (PPSD)**. A PSD and initial procurement plan has been prepared, and both are to be updated during implementation stage based on the capacity and risk assessment of the proposed implementing agencies. Special arrangements like direct contracting; the use of State-Owned Enterprises, UN Agencies, local NGOs, force account or civil servants needs and results-based arrangements may be used and are described in the PSD. The PSD also includes a summary on procurement risk, mitigation action plan, and procurement implementation support and supervision plan. The World Bank has provided support to the project to facilitate the Procurement unit that will be established in the PIU to conduct centralized procurement activities on behalf of the government through preparation of Bidding Documents, issuance of Bidding documents, opening and evaluation of bids, and recommendation of award for the Design, Supply, and Installation any equipment.

79. **Systematic Tracking of Exchanges in Procurement (STEP) and staffing**. The project will use the World Bank’s STEP, an online planning and tracking system that will provide data on procurement activities, establish benchmarks, monitor delays and measure procurement performance. Use of STEP is mandatory for all procurement transactions subject to post and prior review under the project. The project's implementing agencies may have both international procurement consultants and national procurement officers who manage World Bank-financed projects and handle procurement for this project at the national level.

80. **The project design has provided a window for the Recipient to carry out advance contracting and retroactive financing** in accordance with Section V (Paragraphs 5.1 and 5.2) of the World Bank Procurement Regulations for Investment Project Financing (IPF) Borrowers of September 2023.

C. Legal Operational Policies

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	No
Projects in Disputed Area OP 7.60	No

D. Environmental and Social

81. **The key potential environmental and social risks and impacts associated with the project’s activities are related to resource efficiency and pollution prevention and management**, mainly with excavation along side streets and households’ doors, installation of piping and communication infrastructures, water and energy use, dust, noise, and the



potential contamination of water sources, mainly due to construction waste; occupational and community health and safety, as well as human security; potential impacts and risks on habitats or protected areas; impacts and risks of disruption of economic activities and damage to private property; social inclusion of certain categories of marginalized people regarding the development of digital infrastructure and services deployment. The following Environmental and Social Standards (ESS) are applicable in this project: **ESS1, ESS2, ESS3, ESS4, ESS5, ESS6, ESS8 and ESS10**

82. Large-scale infrastructure investments under this nation-wide project, including the deployment of fiber and new cell-sites, may have negative environmental impacts that will need to be proactively managed. Distribution of hardware and digital devices also pose environmental risks related to safe final disposal of hazardous waste. Risks related to e-waste and energy consumption will also require strong mitigation measures that will be done through TA for implementation of e-waste management and by specifying energy efficiency requirements in all procurements undertaken.

83. In addition to environmental impacts, the deployment of digital infrastructure could prompt risks of exclusion. While the project seeks to actively address digital inclusion and expand access to broadband, digital tools and public services, there is a risk of social exclusion of vulnerable or marginalized populations in digital access, such as the elderly or the illiterate, that will need to be actively addressed as part of targeting, especially in relation to access to networks and the various digital skills and device access schemes envisioned. High levels of illiteracy and most people living in rural, often remote areas, add another layer of risk for exclusion from access.

84. Both E&S risks and impacts will be managed during project implementation in accordance with the Environmental and Social Management Framework, the Resettlement Framework, and the Labor Management Plan⁹² consistent with ESS2 requirements. The adopted Environmental and Social Management Framework includes a Security Risk Assessment, an Electronic Waste Management and a Sexual Exploitation and Abuse/Sexual Harassment Action Plan. To ensure consistent and adequate stakeholder involvement throughout the project implementation, an inclusive Stakeholder Engagement Plan⁹³ consistent with ESS10 requirements that identifies the key project stakeholders and the actions to be taken to ensure their inclusion through consultation and participation has been prepared. To further strengthen stakeholder engagement, multi-channel modes of engagement will be integrated into the project, including GEMs, digital tools and analog mechanisms. A negotiated Environmental and Social Commitment Plan was disclosed in the country on August 24, 2024, and on the World Bank website on September 4, 2024.

V. GRIEVANCE REDRESS SERVICES

85. Grievance Redress. Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the World Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the World Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of World Bank non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have been brought directly to the attention of World Bank Management and after Management has been given an opportunity to respond. For information

⁹² Disclosed in-country and on the World Bank website on April 17, 2024.

⁹³ Disclosed in-country and on the World Bank website on April 17, 2024.



on how to submit complaints to the World Bank's GRS, visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the World Bank's AM, visit <https://accountability.worldbank.org>.

VI. KEY RISKS

86. **Overall project risk, after mitigation measures, is rated *Substantial***, based on the following ratings:

- **'Political and Governance' risks are rated *High***. The political situation in Chad has stabilized with the appointment of a new government. Weaknesses in governance frameworks and the lack of institutional capacity pose a challenging backdrop for the reforms the project will support, including broadband market competition. The following mitigation measures are proposed: (i) a PSC was established by the Prime Minister, is chaired by the line Ministry (MCENDA) with the MFBEP as Vice-Chair, and comprises representatives of seven Ministries, the Prime Minister's Office, and the President's Office, to ensure strong and high-level stewardship of the project across different agencies and sectors; and (ii) partnering with regional organizations, and local communities and municipalities to ensure broad support to the project and to assure sustainability of results.
- **'Macroeconomic' risks are rated *High***. The most recent Joint International Monetary Fund–World Bank Debt Sustainability Analysis (January 2023) concluded that Chad is in high risk of debt distress. Chad has a history of commercial borrowing and liquidity constraints that resulted in periods of external and domestic arrears accumulation. The authorities requested debt restructuring under the G20 Common Framework to restore debt sustainability. Chad became the first country to reach an agreement under the G20 Common Framework in November 2022, following the signing of a Memorandum of Understanding by its creditors. The Memorandum of Understanding provides no debt treatment to Chad but commits bilateral creditors to meet any financing gap that may emerge by 2024. The uncertainty regarding future oil prices and production, as well as the volatile security situation, jeopardize progress along the structural reform path. A potential decline in oil prices could result in drastic cuts to public expenditure, risking core development spending, derailing long-term development objectives, and hampering growth. Unpredictable security costs and potential economic disruptions could also divert resources away from important projects. Digitalization of the economy and services will contribute to diversification, economic growth, and domestic resource collection, that will in turn contribute to addressing the debt situation.
- **'Sector Strategies and Policies' risks are rated *Substantial***. While support for the digital agenda has been a government priority, critical policy issues including the government's approach to the USF, the management of government-funded infrastructure, and the weakness of the ICT sector regulator (ARCEP) pose substantial risk. Reforms will involve multiple agencies, potentially diverging agendas and political considerations. The project will place a focus on training and capacity building to ensure that related risks are mitigated.
- **'Technical design of project' risk is rated *Substantial***. The sustainability of project outcomes depends partly on legal and regulatory reforms that should support the project investments. Chad has a limited track-record in reforms of the digital sector and in IPF implementation. Some on-going investments in the digital sector (e.g., Exim Bank-funded data center) do not comply with international standards and pose a risk to project investments in data infrastructure. To mitigate these risks, the project will: (i) coordinate with other World Bank-financed interventions to push the reform agenda; and (ii) focus on high-impact people-centric interventions to limit complexity. Feasibility studies, use of pilots and phased roll-out, as well as AMOA and advisory firms that support the government, will provide an additional layer of quality assurance. Selected services and inclusion of capacity building to ensure knowledge transfer can further limit risks. The IFC could also play a complementary role through its investments; and (iii) implement advisory activities to help the government take digital infrastructure to international standard levels.



- **‘Institutional capacity for implementation and sustainability’ risk is rated *Substantial*.** Experience across the country portfolio suggests that institutional capacity and sustainability is a major challenge. The implementation challenges Chad faces is prevalent in the aspects covered under ANSICE’s mandate. Technical (e.g., overseeing the implementation of the e-signature and trust services regime and auditing cybersecurity compliance) and regulatory (e.g., acting as the Data Protection Authority) difficulties, and ANSICE’s recent move from the Presidency to the Ministry of Security risk further undermining its independence and capacity to implement its mandates. The project will provide TA to develop sustainable business models and systems and place a heavy focus on training and capacity building to ensure that these risks are mitigated.
- **‘Fiduciary’ risk is rated *Substantial*.** Government’s fiduciary capacity is weak. The project will be implemented through a new PIU and led by newly recruited staff. Ownership of the project and the establishment of a new procurement function could pose a significant risk. The centralized procurement at the CEP level, planned with the support of the World Bank fiduciary team, will not only attract international players but also provide technical expertise for the purchase of equipment and services. The World Bank will also provide all necessary support to the CEP in the selection of any study or supervising engineer.
- **‘Environmental and social’ risk is rated *Substantial*.** Large-scale infrastructure investments, including the deployment of fiber and new cell-sites, may have environmental impacts that will need to be proactively managed. Distribution of hardware and digital devices may also pose environmental risks related to safe final disposal of electronic waste. Risks related to e-waste generation and energy consumption will be actively mitigated through TA for implementation of e-waste provisions under Component 1 and by specifying energy efficiency requirements in all procurements undertaken. There are also risks related to potential temporary or permanent displacement due to deployment of broadband infrastructure, risk of exclusion from access to services, especially for populations with low levels of (digital) literacy, and broader risks related to cybersecurity and data protection. To this effect, a Resettlement Policy Framework will be developed. The project will actively seek to address the digital divide (including geographic and gender divides) and address the issue of social inclusion through its focus on rural and underserved areas, targeting of beneficiaries, and adapted sensitization and digital literacy campaigns for trainers, civil servants, and general population. Targeted sensitization campaigns on data protection and cybersecurity are also included.
- **‘Stakeholders’ risk is rated *Substantial*.** Due to its multisectoral focus, the project has a complex stakeholder landscape. Weak inter-ministerial coordination, and the political and donor economy that drives it, could pose a risk to effective project implementation. While considerable consultations have already been undertaken as part of the preparation, which inform the project concept and signal widespread endorsement of key project objectives, additional feedback on design will be sought throughout project preparation. The project is people-centric and will actively seek civic engagement throughout project implementation. World Bank will also continue to provide guidance and advice to the government throughout the process to manage stakeholders and build support for the project.
- **‘Other’ risk associated with security is rated as *Substantial*.** Conflicts and attacks in some parts of the country can affect access to some project sites and security of staff, including implementing agencies, contractors, and World Bank. To help remediate these risks, the E&S impact assessments will outline an SRA to inform project implementation. The project will include flexibility with regard to the selection of sites and consider contingency plans and standard operating procedures to be put in place in case of restricted access. Use of the World Bank GEMS is also expected to help address some of these implementation challenges.



VII. RESULT FRAMEWORK AND MONITORING

PDO Indicators by PDO Outcomes

Baseline	Period 1	Period 2	Period 3	Period 4	Closing Period
Expand access to affordable and resilient broadband connectivity					
Basket prices of broadband services (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
22.50	15	10	8	7	5
People using broadband internet (new use) (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	1,000,000	2,000,000	3,500,000	4,500,000
➤ People using broadband internet (new use) - Female (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	500,000	1,000,000	1,700,000	2,200,000
➤ People using broadband internet (new use) - Youth (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	300,000	500,000	1,200,000	1,800,000
Improve government capacity to deliver public services digitally					
People who made or received a digital payment (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
18	20	22	23	24	25
➤ People who made or received a digital payment (Youth) (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
13	15	17	18	19	20
➤ People who made or received a digital payment (Female) (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
19	21	22	23	24	25
People using digitally enabled services (new services) (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029



0	0	500,000	1,000,000	1,500,000	2,000,000
➤ People using digitally enabled services (new services) - Female (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	300,000	600,000	900,000	1,400,000
➤ People using digitally enabled services (new services) - Youth (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	300,000	600,000	800,000	900,000

Intermediate Indicators by Components

Baseline	Period 1	Period 2	Period 3	Period 4	Closing Period
Component 1. Climate-resilient broadband connectivity and digital inclusion					
Fiber optic deployed (Kilometers)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	45	80	120	160	200
MDAs with new or enhanced broadband by the project (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	5	10	15	30	50
Localities covered in 3G or 4G by the project (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	100	300	400	500	500
Value of private sector investment leveraged under the project (Amount(USD))					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	5,000,000	8,000,000	10,000,000	1,500,000	30,000,000
Beneficiaries completing basic and intermediate digital skills training under the project (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	1,000	2,000	3,000	4,000	5,000
➤ of which female (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	10	20	30	40	50
Chad's eWaste strategy adopted and implemented by GoC (Yes/No)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
No	No	No	No	No	Yes



Reforms adopted (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	1	2	3	4	5
Unique broadband subscribers per 100 inhabitants (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
10	15	20	25	30	35
➤ Share of women (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	10	20	30	40	50
Average price of 1GB mobile data price per month (% of monthly GNI per capita) (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
22.50	20	18	12	8	5
People reporting increased internet use as a result of digital user centers (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	5,000	10,000	20,000	30,000	40,000
➤ Share of women (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	10	20	30	40	50
Development of national universal broadband access-related strategy (Yes/No)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
No	No	No	Yes	Yes	Yes
Component 2. Laying key foundations for digital government services					
Available digitally enabled services (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	5	8	12	16	20
Digital Adoption Index (DAI) (Government cluster) (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
44	45	47	48	49	50
Adults with access to a transaction account (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
22	25	27	30	35	45
National digital and financial literacy Strategy (Yes/No)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029



No	No	No	No	No	Yes
Active users of the digital public platforms supported by the Project (Number)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	750,000	1,250,000	1,500,000	1,750,000	2,000,000
People using payment services (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	200,000	600,000	800,000	1,000,000
➤ People using payment services - Female (Number of people) ^{CRI}					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	100,000	300,000	400,000	600,000
➤ People using payment services - Youth (Number of people)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	0	100,000	300,000	400,000	400,000
Component 3. Project management and knowledge transfer					
Grievances addressed within the project approved timeline (Percentage)					
Sep/2024	Sep/2025	Sep/2026	Sep/2027	Sep/2028	Apr/2029
0	60	70	75	80	90
Component 4. CERC					



Monitoring & Evaluation Plan: Results Indicators

PDO Indicators by PDO Outcomes	
Expand access to affordable and resilient broadband connectivity	
Basket prices of broadband services (Percentage)	
Description	Composite index that measures the cost of a representative set of broadband services. This basket would typically include various broadband plans or packages offered by service providers, and the prices of these services are averaged or aggregated to provide an overall indication of the cost of broadband access in a particular area or for a particular service level.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
People using broadband internet (new use) (Number of people)	
Description	The number of people, public sector facilities, businesses who use new or enhanced internet broadband facilitated through project support. Use is measured as the number of subscribers to broadband internet. (With female and youth disaggregation)
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
Improve government capacity to deliver public services digitally	
People who made or received a digital payment (Percentage)	
Description	Increase access to digital financial services (With female and youth disaggregation)
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.
Responsibility for Data Collection	PIU/MCENDA
People using digitally enabled services (new services) (Number of people)	
Description	The number of people, public sector facilities, businesses who use new or enhanced internet broadband facilitated through project support. Use is measured as the number of subscribers to broadband internet. (With female and youth disaggregation)
Frequency	Annual
Data source	Project implementation reports submitted by PIU



Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
Intermediate Indicators by Components	
Component 1. Climate-resilient broadband connectivity and digital inclusion	
Fiber optic deployed (Kilometers)	
Description	Number of kilometers of fiber optic cables deployed by the project as part of the national backbone.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU, in conjunction with bidders winning tenders.
MDAs with new or enhanced broadband by the project (Number)	
Description	Cumulative number of MDAs with new or enhanced broadband connectivity by the project. MDAs could also include research and innovation agencies, higher education institutions or digital centers hosted in town halls, office posts, community centers (e.g., youth or citizens).
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU, in conjunction with bidders winning tenders.
Localities covered in 3G or 4G by the project (Number)	
Description	Cumulative total number of unconnected villages and cities which become covered with 3G or 4G broadband as a result of the project.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU, in conjunction with bidders winning tenders.
Value of private sector investment leveraged under the project (Amount (USD))	
Description	Private sector investment in digital infrastructure and services leveraged in response to project supported activities.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.



Responsibility for Data Collection	PIU, in conjunction with bidders winning tenders.
Beneficiaries completing basic and intermediate digital skills training under the project (Number)	
Description	Basic and intermediate digital skills training aimed at national education students and teachers especially as part of emergency response through the learning continuity through digital project. These activities may include, but not limited to, learning how to navigate a desktop computer, move and edit files, finding information on the internet, social media, and online safety. (With female disaggregation)
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
Chad's eWaste strategy adopted and implemented by GoC (Yes/No)	
Description	Key metrics of verification are: (a) eWaste strategy adopted by GoC; and (b) Extended Producer Responsibility (EPR) program set-up and operational.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
Reforms adopted (Number)	
Description	This will be the number of regulatory, policy, institutional, administrative and other reforms that are adopted by the government and that impact the adoption and use of broadband internet and digitally enabled services, including the enabling of private investment in these areas and digital related payment reforms.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data consolidated by PIU
Responsibility for Data Collection	PIU
Unique broadband subscribers per 100 inhabitants (Percentage)	
Description	Number of unique mobile broadband subscribers per 100 inhabitants. Mobile broadband subscriptions include 3G, 4G and above (if/when launched) broadband services. (With female and youth disaggregation)
Frequency	Annual
Data source	ARCEP
Methodology for Data Collection	Data will be collected through the statistics report submitted by ARCEP
Responsibility for Data Collection	ARCEP



Average price of 1GB mobile data price per month (% of monthly GNI per capita) (Percentage)	
Description	Cost of purchasing 1GB of mobile data as a proportion of the average monthly income per person in a country. This percentage is used to assess the affordability of mobile data relative to the income of individuals within a country.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by ARCEP and PIU
Responsibility for Data Collection	PIU/ARCEP
People reporting increased internet use as a result of digital user centers (Number)	
Description	This indicator measures the number of people who benefitted from training and capacity-building activities in one of the digital user centers and who reported having increased their internet use thanks to these trainings. This indicator is also disaggregated by female.
Frequency	Annual
Data source	Reports from Digital User Centers
Methodology for Data Collection	Consolidation of annual activity reports from Digital User Centers
Responsibility for Data Collection	PIU
Development of national universal broadband access-related strategy (Yes/No)	
Description	Development of national universal broadband access-related strategy (with a gender lens included)
Frequency	One time
Data source	Joint development and consultations with MCENDA
Methodology for Data Collection	Through the PSC
Responsibility for Data Collection	PIU
Component 2. Laying key foundations for digital government services	
Available digitally enabled services (Number)	
Description	Number of digitally enabled services between government, people, and the private sector (including solutions commercialized by digital companies/startups). This also includes services that are fully or partially digitized (such as digital components of antenatal care services, or blended learning).
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.
Responsibility for Data Collection	PIU/MCENDA
Digital Adoption Index (DAI) (Government cluster) (Percentage)	



Description	Increase availability and adoption of secured and interoperable digital platforms for public services.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.
Responsibility for Data Collection	PIU/MCENDA
Adults with access to a transaction account (Percentage)	
Description	Increase access to digital financial services
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.
Responsibility for Data Collection	PIU/MCENDA
National digital and financial literacy Strategy (Yes/No)	
Description	Development of a gender-smart national digital and financial literacy strategy
Frequency	One time
Data source	Joint development and consultations with MCENDA and MFBEP
Methodology for Data Collection	Through the PSC
Responsibility for Data Collection	PIU
Active users of the digital public platforms supported by the Project (Number)	
Description	Number of individuals who are actively using digital platforms or services that have been developed and deployed through the project.
Frequency	Annual
Data source	Project implementation reports submitted by PIU
Methodology for Data Collection	Data to be consolidated by MCENDA and PIU.
Responsibility for Data Collection	PIU/MCENDA
People using payment services (Number of people) ^{CR1}	
Description	Number of new people using digital payment services, including mobile money (With female desegregation).
Frequency	Annual
Data source	Mobile operators and the Central Bank



Methodology for Data Collection	Reports from mobile money operators and Central Bank
Responsibility for Data Collection	PIU/MCENDA
Component 3. Project management and knowledge transfer	
Grievances addressed within the project approved timeline (Percentage)	
Description	Percentage of grievances or complaints relating to the project, which are addressed within the stipulated service standards for response time. Service standards including response times will be defined in the Project Implementation Manual.
Frequency	Quarterly
Data source	Grievance Redress Mechanism
Methodology for Data Collection	GRM reports administrative data
Responsibility for Data Collection	PIU/MCENDA

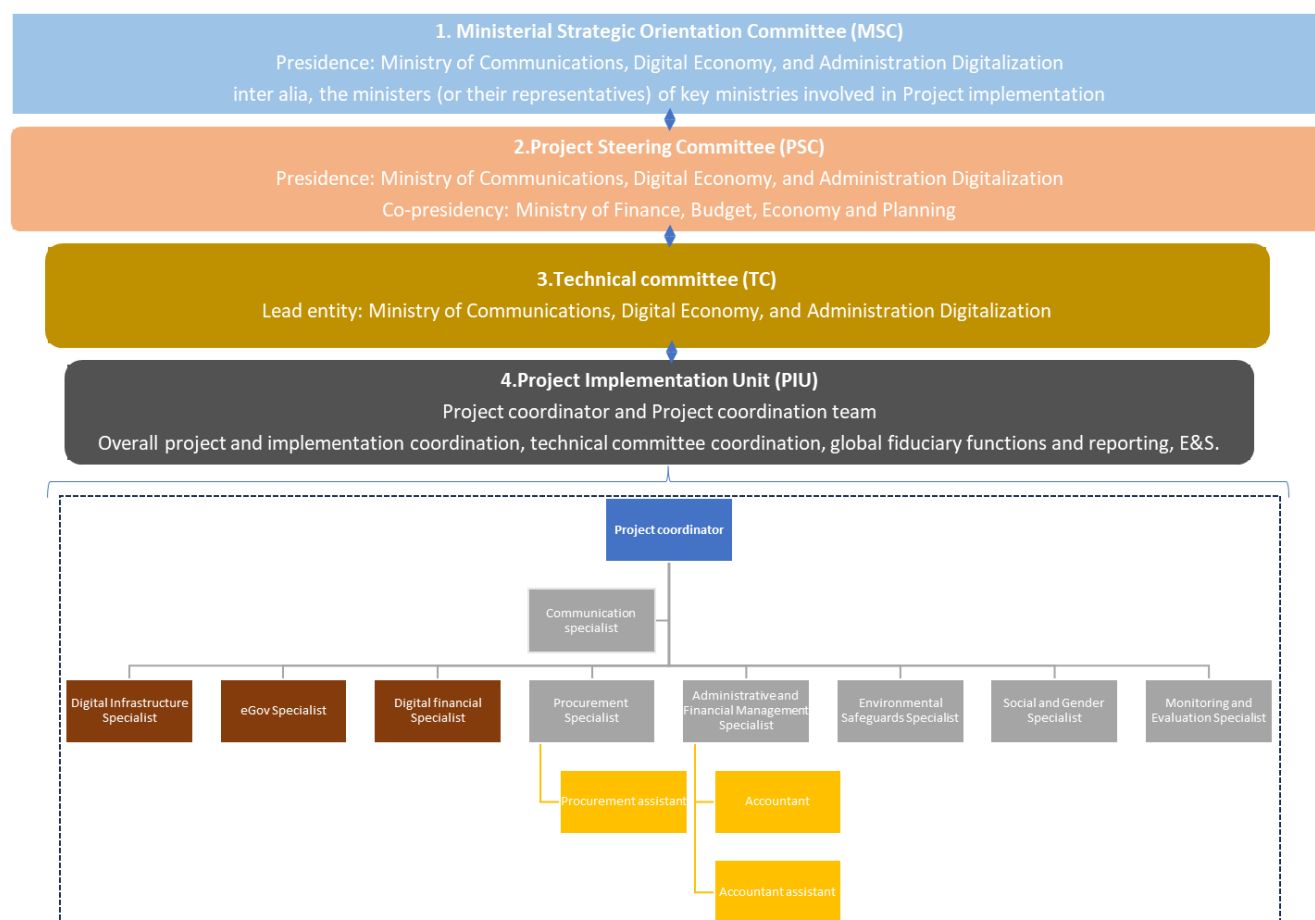


ANNEX 1: Implementation Arrangements and Support Plan

A. Institutional Arrangements

1. **The MCENDA will serve as the implementing agency for the entire project.** The PIU under the MCENDA will have the responsibility of the day-to-day management of the project and will be responsible for the implementation of all project activities and the fiduciary aspects of the project. The Ministerial Strategic Orientation Committee chaired by MCENDA and which shall include, inter alia, the ministers (or their representatives) of key ministries involved in Project implementation; and which shall be responsible to oversee the general direction of the project, providing strategic guidance and ensuring consistency with national development policies. PSC chaired by the Secretary General of the Ministry of Finance, Budget, Economy and Planning and which includes representatives of the Recipient's key ministries involved Project implementation and responsible for providing overall guidance, strategic decision-making, approving the Annual Work Plan and Budget, and overall oversight and coordination of project implementation. The PIU and Technical Committee related to project implementation will report to the PSC on a quarterly basis or at the request of the PSC.

Figure 1.1 Project Institutional Arrangements





B. Financial Risk Assessment and Financial Management Action Plan

2. **The MCENDA-PIU will be the World Bank's main counterpart and focal point for FM aspects of the project.** This includes budgeting, financial reporting, supervision, management of the Designated Account, and auditing. Overall responsibility for preparing an annual work plan and related budget will lie with the PIU. The different steps of budget elaboration and management (preparation, revision, adoption, and execution) will be detailed in the FM section of the PIM. The annual work plan and budget will be prepared annually for approval by the PSC and submitted to the World Bank for approval for no-objection not later than November 30 of each year proceeding the year the work plan should be implemented. A budget execution report will be included in the reporting scheme to enable the monitoring of the project implementation.

3. **The FM functions will be carried out by a team** including: (i) a qualified and experienced FM specialist; (ii) a qualified and experienced accountant; and (iii) an accounting assistant to be recruited through a competitive process in compliance with World Bank's rules. The PIU's FM team will assume the overall responsibility for maintaining accounts associated with project's activities and ensuring that annual financial statements are produced in a timely manner, and in accordance with accounting standards. The PIU will use SYSCOHADA accounting standards related to non-profit entities (SYSCEBNL) which is commonly used amongst West and Central African Francophone countries. This will be ensured through the purchase and customization of the accounting software used for the World Bank-financed projects in Chad, that will be parameterized accordingly. The software might be capable of keeping records of the project financial activities and generating financial statements that are in line with the project reporting requirements. The software might be able to also record procurement transactions and keep track of the project assets.

4. **The PIM including the FM manual will provide guidance on project implementation arrangements**, and notably on roles and responsibilities within the MCENDA-PIU, as well as on the technical, administrative, financial, and accounting procedures, procurement arrangements, and the safeguard procedures adopted by the project implementing unit, in accordance with the respective national laws and regulations. The PIU will be responsible for maintaining all necessary controls to ensure that: (i) funds are used for the purposes for which they were intended in an efficient and economical manner; (ii) transactions are accurately and completely recorded; (iii) financial reports are prepared and submitted on time; and (iv) project assets are well protected.

5. **An experienced internal auditor will be recruited** within three months after project effectiveness. Transparency, accountability, and anti-corruption efforts will be encouraged including via a complaint handling mechanism; a communication strategy to inform the public through the media on all aspects of the project; and the publication on the PIU or government websites of budgets, financial reports and audited financial statements. The PIU will also have to deal with fraud and anti-corruption in accordance with the World Bank Anti-Corruption Guidelines referred to in the Financial Agreement.

C. Flow of Funds and Disbursements Arrangements

6. **The disbursement methods to be used under this project will be based on the Disbursement Guidelines for IPF**, issued on May 4, 2017. Upon effectiveness, this operation will follow the transaction-based disbursement method. Direct payment, reimbursement, and special commitment methods will be available to the project and might apply as appropriate. Further details about disbursements to the project will be included in the disbursement procedures described in the Disbursement and Financial Information Letter (DFIL) and the PIM.



7. **Banking arrangements for eligible expenditures.** The PIU will open a designated account (DA) denominated in Central African CFA franc (XAF) in a commercial bank on terms and conditions acceptable to IDA for payment of eligible expenditures. The project's DA will function under the co-signature of the project Coordinator and the FM Specialist of the PIU. If ineligible expenditures are found to have been made from either of the DAs, the Recipient will be obligated to refund the same. If the DA remains inactive for more than six months, the Recipient may be requested to refund to IDA amounts advanced to the DAs. IDA will have the right, as reflected in the Financing Agreement, to suspend disbursement of funds if reporting requirements are not complied with. IDA will make an initial advance disbursement in Central African CFA franc (XAF) into the DA opened for this purpose, upon receiving a withdrawal application from the Recipient. Replenishment of funds from IDA to the DA will be made upon evidence of satisfactory utilization of the advance, reflected in statement of expenditures replenishment applications would be required to be submitted regularly monthly. Further details about disbursements to the project will be included in the disbursement procedures described in the DFIL. Fund flow arrangements between various implementing entities and reporting requirements will be detailed in the PIM.

Table 1.1. Categories of Eligible Expenditure

Category	Amount of the Financing Allocated (expressed in SDR)	Percentage of Expenditures to be Financed (inclusive of Taxes)
(1) Goods, works, non-consulting services, and consulting services, Training and Operating Costs for the project except Subsidies under Part 1.2(b) of the project	35,475,000	100%
(2) Subsidies under Part 1.2(b) of the project	30,150,000	100%
(3) Emergency Expenditures under Part 4 of the project	0	
(4) Refund of Preparation Advance	3,875,000	Amount payable pursuant to Section 2.07 (a) of the General Conditions
TOTAL AMOUNT	69,500,000	

8. **Financial Reporting Arrangements.** The MCENDA-PIU will prepare quarterly unaudited IFRs in form and content satisfactory to the World Bank, which will be submitted to the World Bank within 45 days after the end of the calendar quarter to which they relate. The contents of the IFR will include the following information to account for Project funds:

- Statement of Sources and Uses of Funds.
- Statement of Uses of Funds by Project Activity/Component.
- DA Activity Statement.
- Bank statements for the DA and related bank reconciliation statements.
- Summary statement of DA expenditures for contracts subject to prior review.
- Summary statement of DA expenditures not subject to prior review.

9. The PIU will also prepare annual financial statements for the project within three months after the end of the accounting year, and these statements will comply with respective accounting standards and World Bank requirements. The audited financial statements will be required to be submitted to the World Bank within six months after the end of the fiscal year.



10. **External Audit Arrangements.** The annual financial statements prepared by the PIU as well as the internal control system will be subject to an annual audit by a reputable and independent auditing firm based on ToR satisfactory to the World Bank. The project will meet the cost of hiring a private audit firm. The audit will be done in accordance with International Standards on Auditing. The external auditors must be appointed within six months of the project's effectiveness date. The audit report for the project accounts, together with the management letter, should be submitted to the World Bank within six months after the end of the government's fiscal year (December 31). The audit reports will be publicly disclosed by the World Bank in accordance with the World Bank disclosure policy. In addition, in accordance with the World Bank Policy on Access to Information, the World Bank requires the PIU to disclose the audited financial statements in a manner acceptable to the World Bank.

11. **The FM Action Plan** described below has been developed to mitigate the overall FM risks.

Table 1.2. FM Action Plan

	Issue	Remedial action recommended	Responsible entity	Due date
1.	Staffing	Recruit a FM specialist and an accountant, based on ToR satisfactory to the World Bank	MCENDA/ PIU	Before effectiveness
2.	Internal audit	Recruit, based on ToR satisfactory to World Bank, an experienced internal auditor	MCENDA/ PIU	Within three months of the effectiveness
3.	FM manual of procedures	Elaborate and adopt the PIM including detailed FM procedures	MCENDA/ PIU	Before effectiveness
4.	Information system accounting software	Acquire and install accounting software	MCENDA/ PIU	Within three months after the effectiveness
5.	External auditing	Recruit an independent auditor with qualifications acceptable to the World Bank	MCENDA/ PIU	Within six months after effectiveness (recruited to audit three years of accounts)

12. **Implementation Support Plan:** Based on the outcome of the FM risk assessment, the following implementation support plan is proposed. The objective of the implementation support plan is to ensure the project maintains a satisfactory financial management system throughout the project's life.

Table 1.3. FM Implementation Support Plan

FM Activity	Frequency
Desk reviews	
Interim financial reports review	Quarterly
Audit report review of the project	Annually
Review of other relevant information such as interim internal control systems reports.	Continuous as they become available
On site visits	



FM Activity	Frequency
Review of overall operation of the FM system.	Twice per year (Implementation Support Mission). Once per year as soon as the FM residual risk becomes Moderate.
Monitoring of actions taken on issues highlighted in audit reports, auditors' management letters, internal audit, and other reports.	As needed
Transaction reviews (if needed)	As needed
Capacity building support	
FM training sessions	During implementation and as and when needed.

D. Procurement

13. **Procurement under the proposed project will be carried out in accordance with the World Bank “Procurement Regulations for Investment Project Financing Borrowers”** dated September 2023, the ‘Guidelines on Preventing and Combating Fraud and Corruption in Projects Financed by IBRD Loans and IDA Credits and Grants’ dated October 15, 2006, and revised in January 2011 and as of July 1, 2016, and beneficiary disclosure requirements, as well as other provisions stipulated in the project Legal Agreements.

14. **Project Procurement Strategy for Development (PPSD).** An assessment of client’s capacity has been conducted and summarized below. The detailed PPSP and the procurement plan have been developed based on the capacity and risk assessment of the proposed implementing agencies. The PPSP describes how procurement activities will support project operations for the achievement of project development objectives and deliver Value for Money (VfM). The procurement strategy is linked to the project implementation approaches ensuring proper sequencing of the activities. Special arrangements like simplified procurement process in accordance with World Bank’s Procurement Regulations, direct contracting, the use of Statements of Expenditure, UN Agencies, force account, and results-based arrangements may be used, and are described in the PPSP. The PPSP also includes a summary on procurement risks, mitigation action plan, and procurement implementation support and supervision plan.

15. **Systematic Tracking of Exchanges in Procurement (STEP).** The project will use the World Bank’s STEP, an online planning and tracking system that will provide data on procurement activities, establish benchmarks, monitor delays and measure procurement performance. Use of STEP is mandatory for all procurement transactions subject to post and prior review under the project.

16. **Procurement Staffing Capacity.** The Implementing Agency will have a senior procurement consultant and national procurement officers who will manage the Bank-financed project activities and handle procurement for the project at national and international level.

17. **Procurement Planning, retroactive financing, and Standard Procurement Documents.** A first Procurement Plan, which is the output of the PPSP, outlines the procurement procedures to be used. Any major procurement packages as identified in the PPSP will be done by the PIU. The Procurement Plan details the activities to be carried out during the first 18 months is prepared. The project design has provided a window for the Recipient to carry out advance contracting and retroactive financing in accordance with Section V (Paragraphs 5.1 and 5.2) of the World Bank Procurement Regulations for IPF Borrowers of September 2023. The project shall use the World Bank’s latest standard procurement document versions for the procurement of Consulting Services, Non-Consulting Services, and Goods.



18. **Procurement Post Reviews (PPRs) and Independent Post Reviews (IPRs) by the World Bank.** Based on the assessed agency implementation risk for procurement, which is Substantial Risk, the Bank will carry out PPRs or IPRs for all contracts based on the approved procurement plan not subject of prior review by the Bank.

19. **Procurement Risk.** The overall procurement risks for the project are significant; this rating is due, among other things, to: (i) low capacity of the actors involved in carrying out the processes; (ii) inefficiency and delays in the procurement process; (iii) insufficient competition in procurement process; (iv) a weak recourse system; and (v) collusion and interference in the procurement process. The procurement risk associated with this innovative project may be different from that of other projects in the portfolio. As part of the World Bank's support to the implementation of government-owned projects, procurement risks and the performance of the ministries have been regularly monitored and evaluated. Over the last five years, the procurement rating has moved from acceptable to poor.

Table 1.4. Country specific procurement risks and mitigation measures are in tables below:

Risk/Issue	Action	Resp.	Timeline
Staff involved in the Project may not have enough knowledge of the New Procurement Framework and/or risk of confusion with the former guidelines.	*Organize workshop sessions on the New Procurement Framework to train all staff involved in the procurement of the Project and Train all the new members of the Ministry tender committee as well as staff at others and stakeholders	PIU	Within two months of effectiveness
Delays in (i) bidding process due to delay in preparation of technical inputs, and (ii) approval of bid evaluation reports.	*Put in place (a) mechanisms to anticipate the preparation of technical inputs in bidding documents; * Strict compliance with the legal deadlines set for each stage of the procurement process	PIU	Within two months after effectiveness
Delays in contracts execution	*Contract management plans for major contracts and close monitoring	PIU	Throughout project implementation
Inadequate communication and interaction between the beneficiaries and the PIU, which may lead to delays in procurement processes and poor estimation of the costs	*Improve communication through regular meetings *Engage subject matter technical specialists in the entire procurement cycle of procurement activities.	PIU	Within six months after effectiveness
Procurement in a specialized market in a fragile area with few bidders can restrict competition and possibly increase prices and collusion risks	*Organize procurement red flags training in collaboration with Integrity Vice Presidency (Preventive) for the PIU.	PIU World Bank	Within three months after effectiveness
Poor contract management and administration of big contracts	*Develop contract management plans for prior review. Keep large value works on World Bank's radar for regular follow-ups with support from World Bank technical and Environmental and Social Framework team members, as may be needed for review.	PIU World Bank	Within two months after effectiveness
Poor filing which can lead to loss of documents	*Put in place a filing system at the PIU to ensure compliance with World Bank procurement filing manual.	PIU	Throughout project implementation



ANNEX 2: Contribution to IDA20 Commitments

	Objective	Relevant Policy Commitment	Contribution of the proposed operation
Jobs and Economic Transformation			
1	Expanding broadband access and usage for jobs of the future	To close the connectivity gap, IDA will support 17 IDA countries, including those which will benefit from IFC's support under the IDA PSW to develop digital infrastructure, increase inclusive, secure, and affordable access to and usage of broadband connectivity, among which are six landlocked countries, four Small States and nine countries in fragile and conflict-affected situations.	Sub-component 1.1 will support the GoC in creating a conducive legal and regulatory environment for the telecom/broadband sector, which is expected to result in improved broadband affordability and increased internet penetration throughout the country (including among women). Component 2 will expand broadband access across the country under an MFD approach – both in (i) connecting MDA to broadband in urban and peri-urban areas; and (ii) last-mile access networks in underserved / uncovered rural areas.
2	Positioning more firms for recovery, including through the adoption of digital technology	Support programs in 15 IDA countries, to strengthen private sector recovery and transformation that are well targeted, inclusive of small and medium enterprises and support the adoption of digital technologies, with monitoring to capture distributional impacts and effectiveness.	Sub-component 3.1 will strengthen the national ICT school that would help increase the production of new ICT talents and professionals in the job market.
Gender and Development			
3	Closing the gap in digital technology	At least 30 IDA20 operations in digital development, financial inclusion, and agriculture will increase women's access to and usage of digital technology to close gender gaps in access and usage.	Sub-components 1.3, 2.3 and 3.1 will support women's increased access to and usage of digital services. All activities will be gender-sensitive, and gender-disaggregated indicators will be monitored. The project will also conduct a gender impact evaluation, assessing the impact of one or more sub interventions aimed at digitally empowering women.
Human Capital			
4	Expanding adaptive social protection and building resilience to shocks	To ensure inclusive and effective response against shocks and crises, support at least 20 IDA countries' resilience by building adaptive social protection systems, including the use of digital technologies.	Investments under sub-components 3.1 and 3.3 will support improvements in government back-end systems to support inter alia the social protection sector through use of digital payments and interoperability between systems, thus improving access to and delivery of social protection services to beneficiaries, reducing costs for the government and increasing the country's resilience to shocks and crises.
5	Expanding access to core services for persons with disabilities	To promote inclusive societies, support at least 18 IDA countries to meet the needs of persons with disabilities by implementing the principles of universal access through projects in education, health,	Sub-components 1.3, 2.3, 3.2 and 3.4 will support the development of digital public services that are universally accessible, which are designed in a manner that can



	Objective	Relevant Policy Commitment	Contribution of the proposed operation
		social protection, water, urban, digital development and/or transport.	be accessed, understood, and used by all persons regardless of gender, disability, age, use of assistive devices, location or internet access.
Governance and Institutions			
6	Enabling digital government services	Support at least 20 IDA countries to adopt universally accessible GovTech policies, regulations, or solutions to enable digital government services.	Sub-components 1.2 and 3.1 will build a trusted environment for development and use of digital services and strengthen key entities' institutional and technical capacity for the implementation of digital regulation and governance. Sub-component 2.2 by connecting selected MDAs and sub-component 3.1 building digital public infrastructure, will enable the delivery of digital public services in key/essential sectors (social and productive) and contribute to the public sector transformation.



ANNEX 3: Gender Analysis, Proposed Actions, and Relevant Indicators

Barriers/Gaps	Strategic Actions and Project Design Decisions	Indicators
Component 1. Climate-resilient broadband connectivity and digital inclusion		
Sub-component 1.1. Strategic, policy, legal, regulatory and institutional frameworks for resilient and affordable broadband access		
Despite Chad adopting Law N°014/PR/2014, which sets the requirements for the establishment of a Universal Electronic Communications Service Fund, women lag behind men in access and use of the Internet. The high gender gap in Internet use can, in part, be attributed to a lack of implementation of key national and regional frameworks, including the aforementioned law, which aims to close the gender digital divide by making the Internet cheaper and easier to access for those who need it most, including women and girls.	Support MCENDA, ADETIC, and ARCEP in: (i) undertaking a gender-smart analysis of the legal and regulatory framework and support of digital development; (ii) enabling the collection of data from mobile network operators in respect to women subscription levels and services usage; (iii) developing a broadband policy with targeted strategies to expand coverage to women and girls in rural areas; and (iv) undertaking an assessment of the intersectional barriers (including age, socioeconomic status, geographic location, disability status, etc.) of adoption of digital services (including broadband services) to better understand key demand-side and gender-specific user constraints in order to adequately address them.	- Development of national universal broadband access-related strategy (with a gender lens included) (Yes/No) <i>Note: Disbursement interval to be determined in coordination with the client, e.g., quarterly or bi-annually.</i>
Sub-component 1.4 Digital inclusion		
In 2022, the gap in adult literacy between men and women stood at 16.9 (in favor of men) and was consequently larger than the gap of the Sub-Saharan Africa aggregate (which stands at 12.7). ⁹⁴ Furthermore, Chad is particularly characterized by women's low enrollment in STEM fields and ICT degree programs. It is reported that women's tertiary enrollment is as low as 1.5 percent (versus 5 percent for men) while women only account for 4.8 percent of the researchers. In 2019,	To support women and girls in acquiring digital skills, these targeted activities will (i) deliver climate-oriented digital skills training programs (including digital literacy, financial literacy, cybersecurity, data protection, consumer protection trainings and digital technologies that can be leveraged for climate adaptation and protection trainings and digital technologies that can be leveraged for climate adaptation and mitigation—e.g., early warning systems, digital cash transfers in the event of shocks, solar-powered devices, energy efficient equipment, etc.) to equip young trainers to enhance basic and intermediate	- Beneficiaries completing basic and intermediate digital skills digital literacy training programs under the project (Number), of which female (%). Baseline: 0; Target: 50%. - People reporting increased internet use as

⁹⁴ UNESCO Institute for Statistics (UIS), 2023.



Barriers/Gaps	Strategic Actions and Project Design Decisions	Indicators
less than 1 percent of women reported having any digital skills, while only 3.4 percent of men reported having at least one out of nine digital skills measured by the Multiple Indicator Cluster Survey. Women and girls tend to be at a disadvantage in both foundational skills at an early age (among and 7–14-year-olds) and digital skills, between the age of 15 and 24. Young women and girls experience substantial barriers in terms of developing their digital skills, including very basic digital skills such as copying or pasting files or folders, sending emails, or transferring files. Consequently, women participate in the labor market at a lower rate than men (64.4 versus 77 percent) and are under-represented in the digital economy. Additionally, there is a growing digital gender divide in Internet access and use, as 0.7 percent of women (versus 6.7 percent of men) report using the internet on a regular basis.	digital skills for internet adoption; (ii) implement targeted outreach and initiatives for women and girls, including dedicated digital ambassador programs to teach rural communities how to use digital tools tailored to local needs; (iii) develop targeted content and approaches (within training communities) designed to close observed gender gaps in access to digital devices and digital skills; (iv) leverage female role models within the sector to build women’s self-efficacy; and (v) Initiate men’s engagement programming and design couples’ interventions to ease restrictive social norms, thereby promoting women’s uptake of training programs and enhancing their access to and use of digital devices. TA will also support the set-up of additional ‘digital user centers’ conducive to women’s use and comfort by specifically ensuring adequate coverage in remote and poor locations where women are often concentrated, creating a safe environment through tailored delivery modalities (including female trainers and intermediaries available during the times of the day that women would generally make use of the digital user centers), and opening times to account for women’s time constraints and leisure time. Set up of these centers will include interventions to support women and girls’ online safety and incorporate anti-discrimination and anti-sexual harassment policies and reporting mechanisms.	a result of digital user centers (Number); Share of women Baseline: 0; Target: 50%
Component 2. Laying key foundations for digital government services		
Component 2.3. Digitization of payments		
Key policy documents relating to consumer protection and financial inclusion (such as the 2016 National Financial Inclusion strategy) are out-of-date and lack a gender lens to reduce the barriers women face to benefit from DFS. As such, Gender disparities are observed in digital financial inclusion with only 13 percent of women as opposed to 15 percent of men having used a mobile phone or the internet to send money in 2022. Several constraints limit women's access to DFS,	Activities will support the development of a National Digital and Financial Literacy Strategy, designed to address gender-specific barriers to women’s digital inclusion and implement related programs, by (i) setting clear measurable targets for financial inclusion outcomes; (ii) establishing clear accountability structures for achieving targets; (iii) consulting women groups and experts in the policy development process; (iv) building awareness and benefits of DFS for women, through digital	- Development of a gender-smart National digital and financial literacy Strategy (Yes/No)



Barriers/Gaps	Strategic Actions and Project Design Decisions	Indicators
which include sociodemographic and cultural factors contributing to women's lower income per capita, limited eligibility financing, and lower participation. in training and initiatives in financial education. There is scope for improvement on gender equality within the national legal framework, particularly as it relates to specific targets and closing digital gender gaps in access, usage, DFS and skills.	skills and financial literacy programs (e.g., for social payments recipients); (v) disaggregating on access and use of digital financial services; and (vi) integrating online safety through training to address cyber risks. The project will also consider complementary interventions designed to close DFS and financial inclusion gender gaps, including digital platforms specifically designed for low-literacy populations, such as image-based mobile-money agent applications that increase the transaction transparency for low-literacy clients as well as coupling digital skills training with financial literacy training for women.	